

IBM Resiliency Summit

Continuity of Operations: GDPS & CyberVault

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Agenda

- GDPS Overview
- IBM Z Cyber Vault Overview
- What Are Financial Services Enterprises Doing?
- Best Practices
- Summary

GDPS Level-set

IBM GDPS is installed in several of the largest enterprises in the world.

83% of the top **40** banks around-the-world use GDPS

Over **25+** years of experience

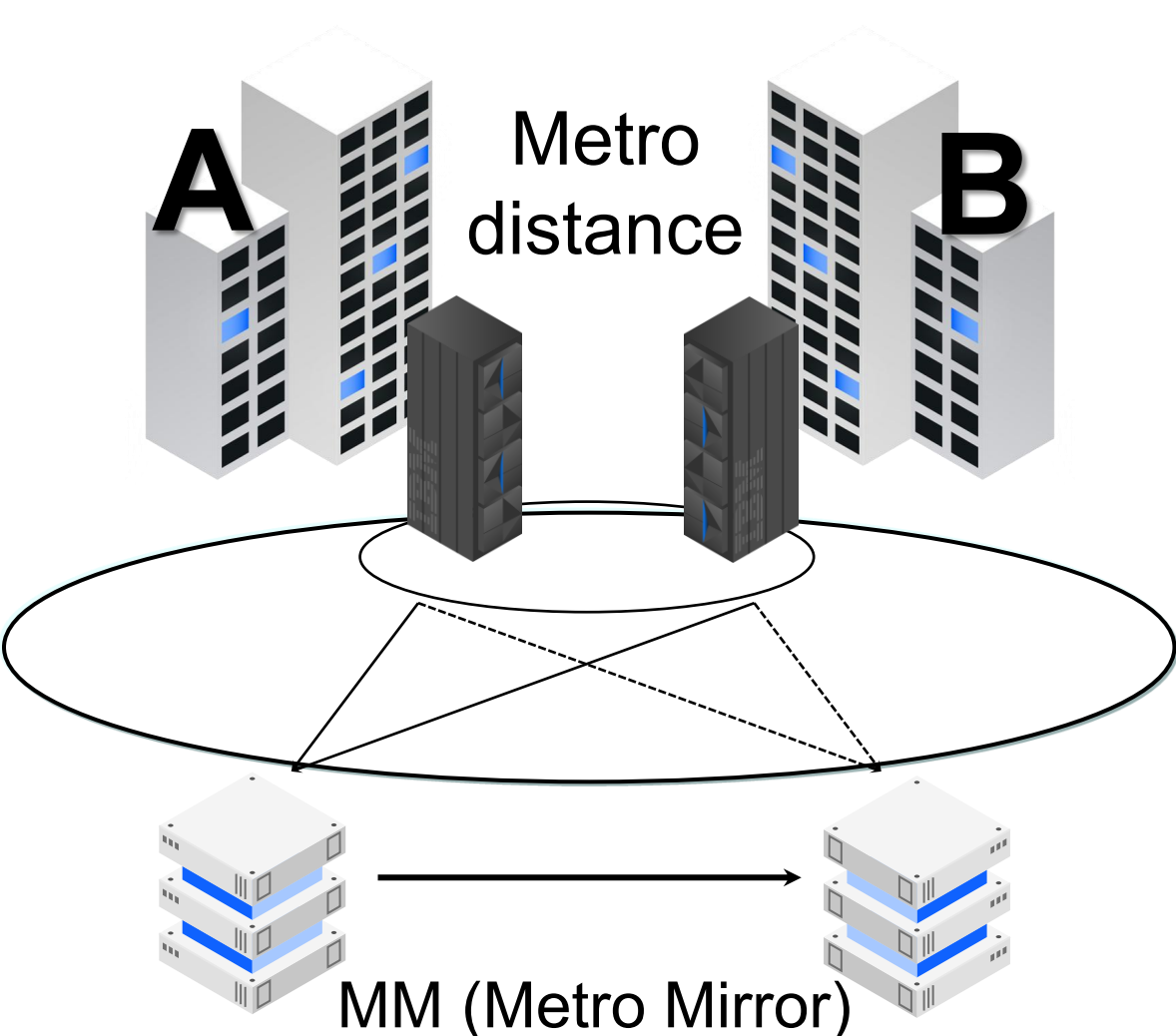
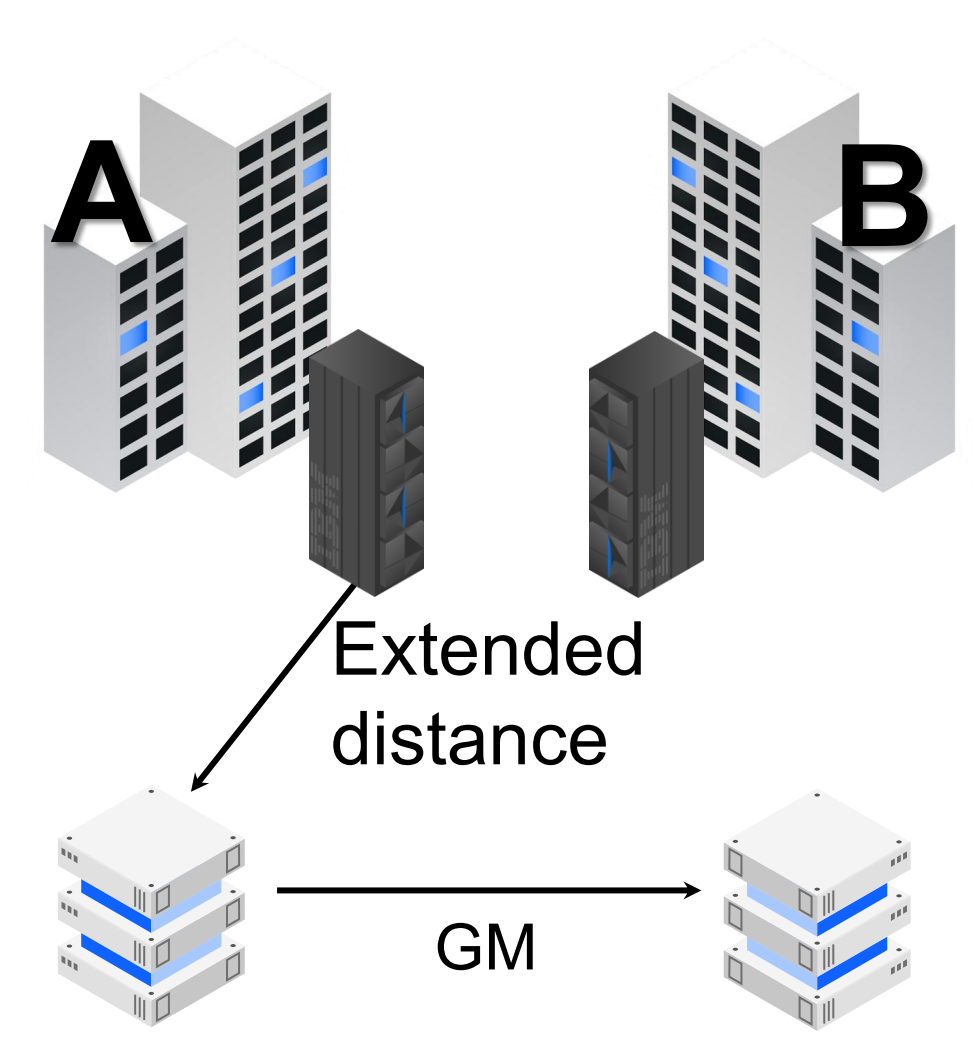
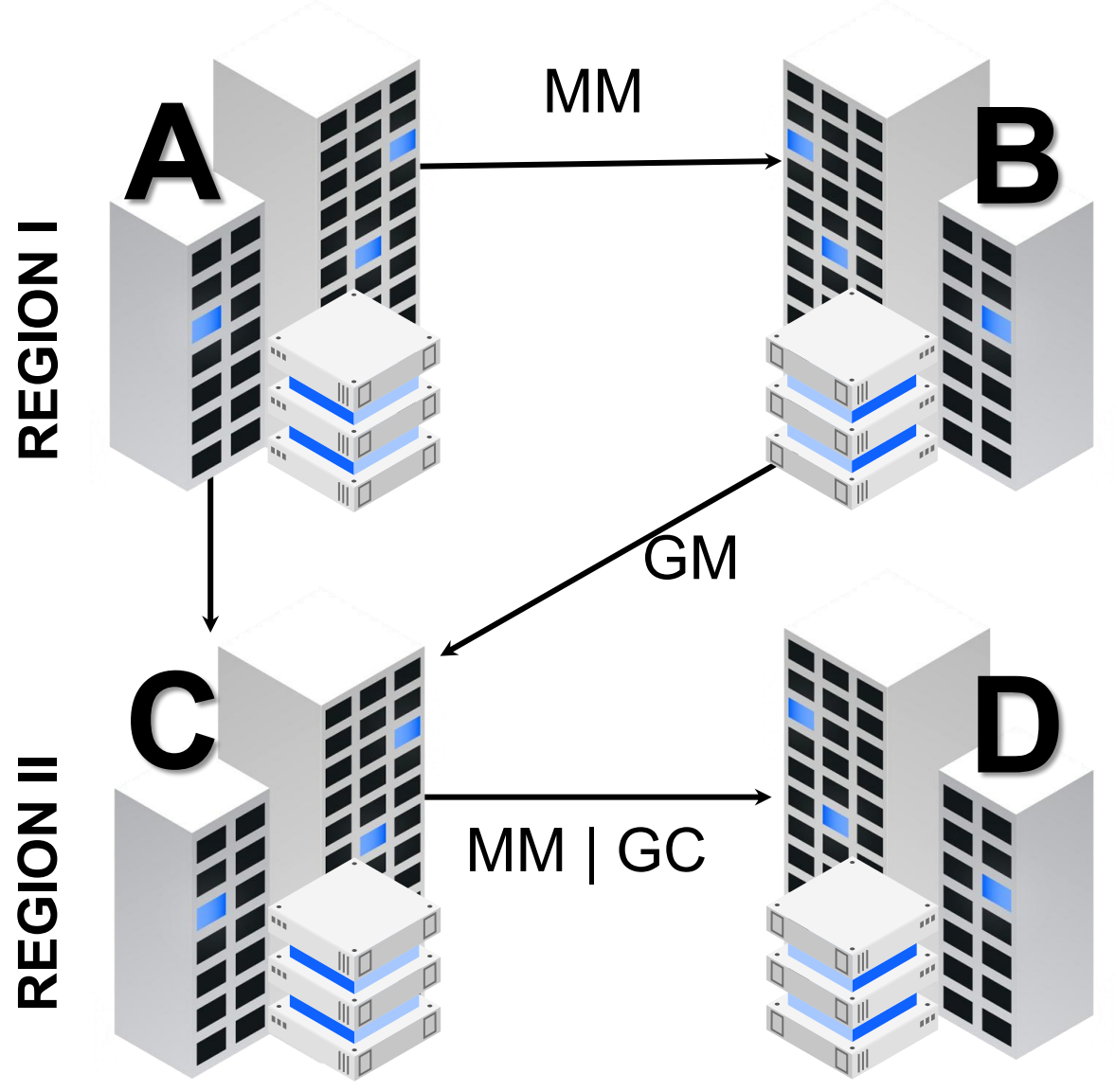
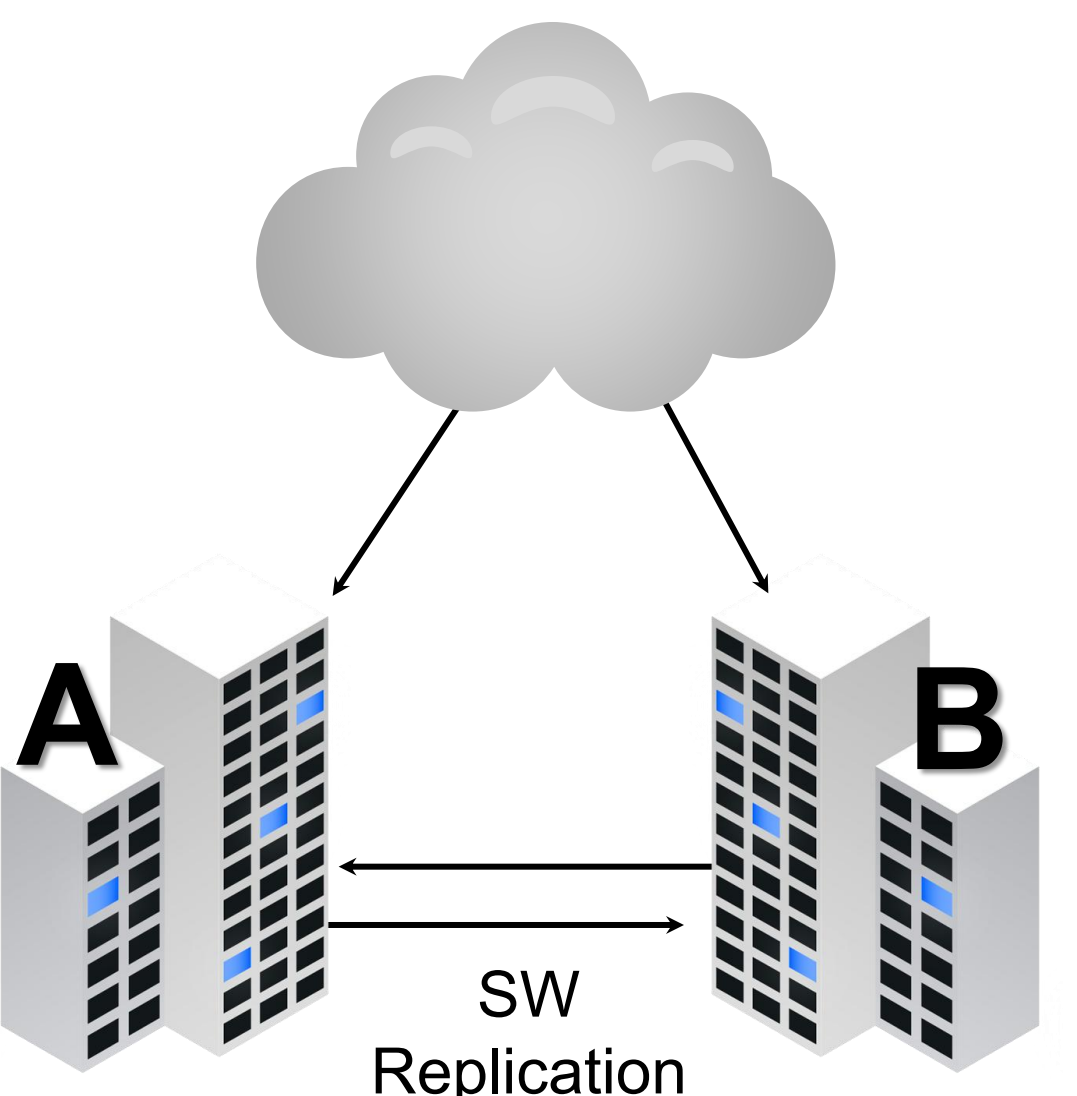
+99% of the new GDPS deployments are with DS8K

50+ people work on GDPS

2+ MLOC in GDPS



GDPS core topologies

GDPS Metro	GDPS Global	GDPS Metro Global	GDPS Continuous Availability
Near-continuous availability and recovery at metro distances	Disaster recovery at extended distance	Near-continuous availability regionally & recovery for 3,4 & 6 sites	Near-continuous availability, recovery & workload balancing
Systems remain active Multisite workloads can withstand site and storage failures 	Rapid systems DR with “seconds” of data loss 	Metro near-continuous availability and out of region disaster recover 	Continuous availability at unlimited distances 

GDPS license dynamics YE 2025

1,300+

GDPS licenses now in use worldwide

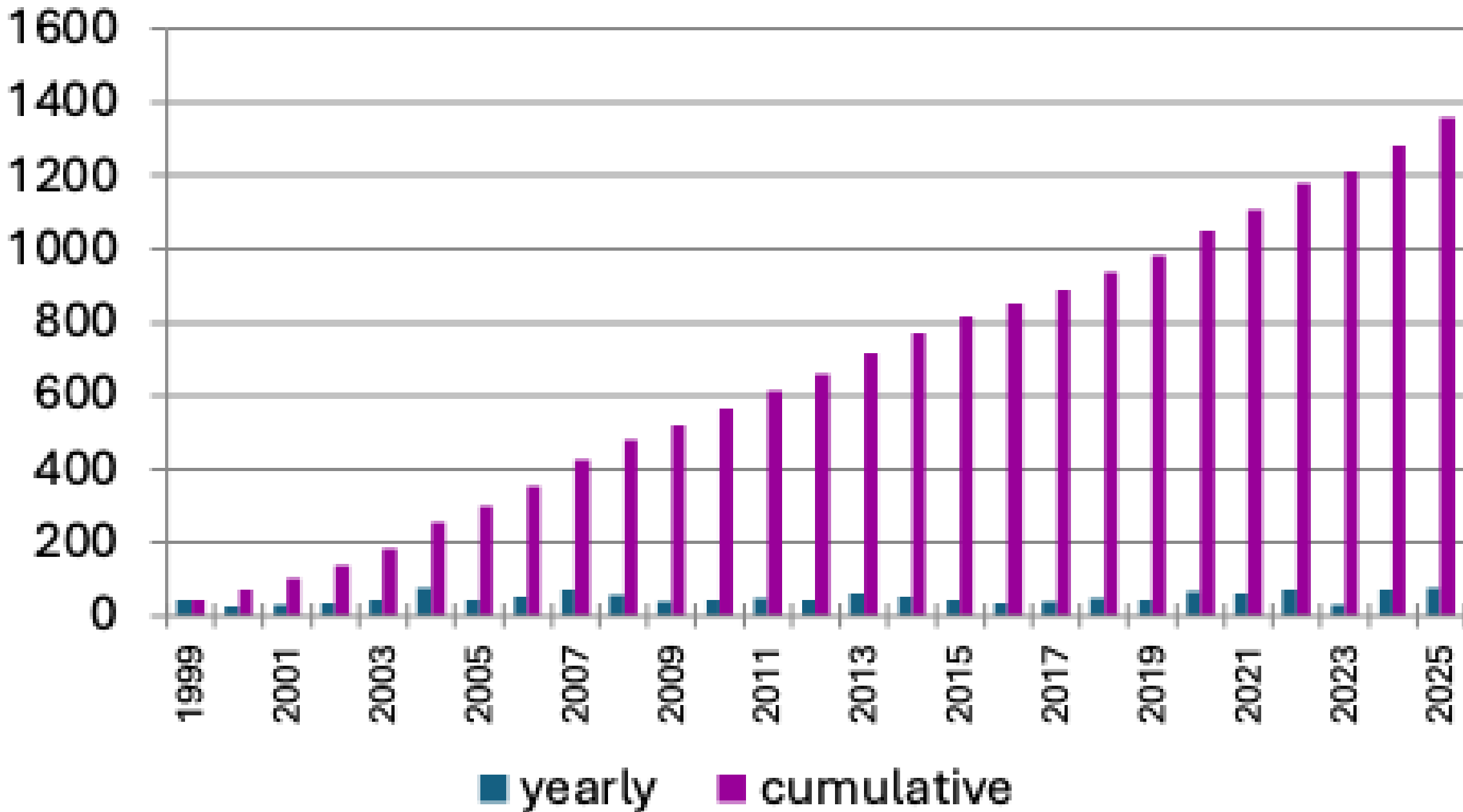
+14%

Growth in MGM4 environments
2024 to 2025

84%

Of the worlds top 25 banks are using GDPS

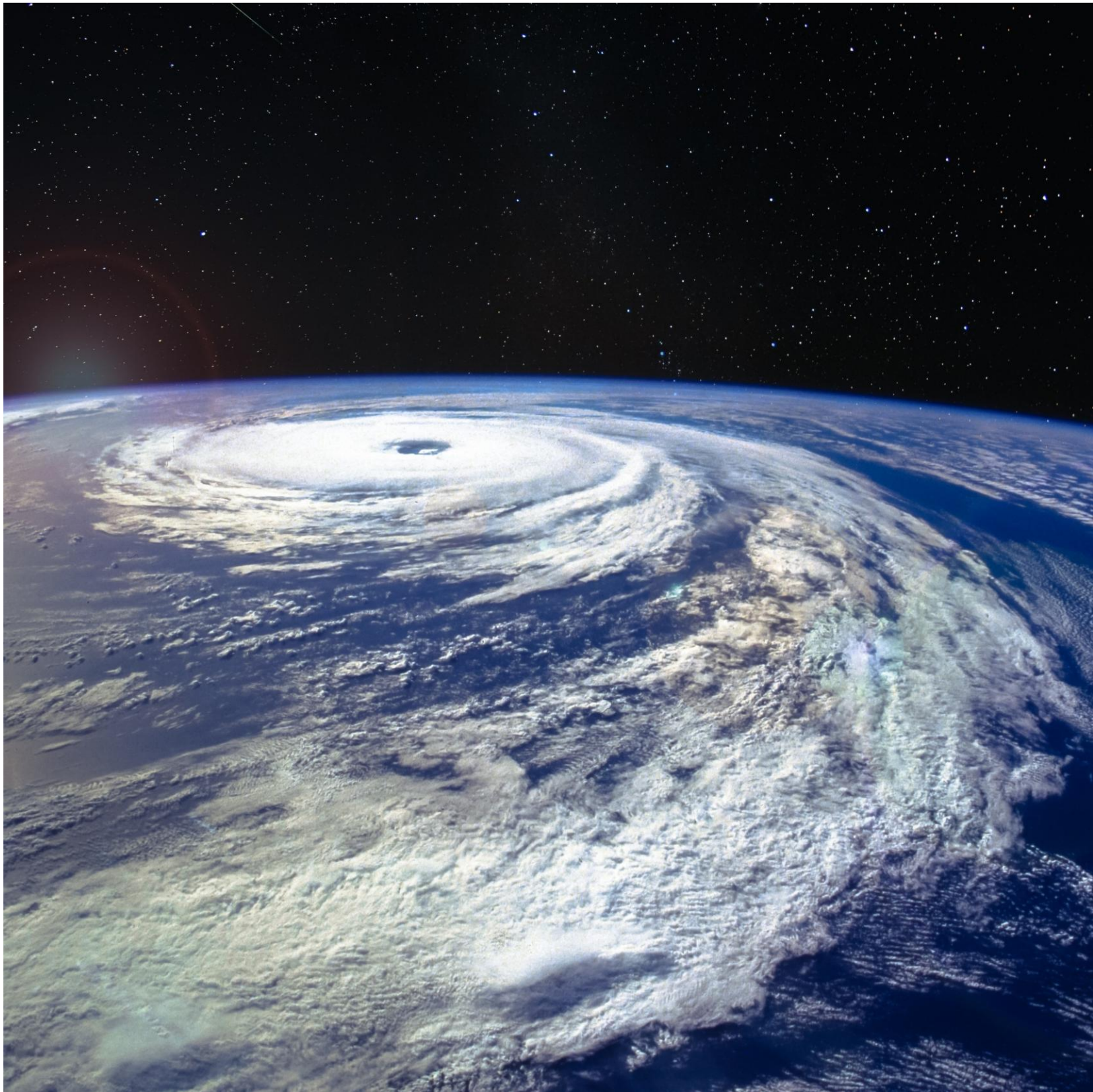
GDPS Licences



Multi site configuration	Environments	Clients
MGM3	125	65
MGM4	80	29

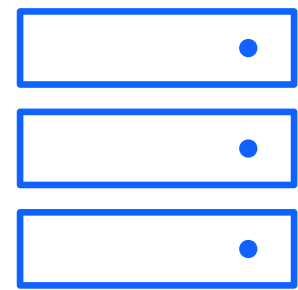
IBM Z Cyber Vault Overview

Traditional resiliency solutions will not protect you from cyber attack



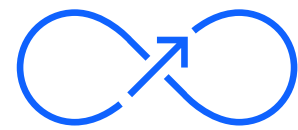
	Traditional resiliency for HA/DR	What's required for Cyber Resiliency
Replication	Data is being replicated continuously but logical errors are also replicated instantaneously	Scheduled point in time copies stored in an isolated, secure location
Error detection	Immediate detection of system and application outages	Regular data analytics on point in time copies to validate data consistency
Recovery points	Single recovery point that likely will be compromised	Multiple recovery points
Isolation	All systems, storage and tape pools participate in the same logical system structure	Air gapped systems and storage so that logical errors and malicious intruders can not propagate
Recovery Scope	Continuous availability and disaster recovery	Forensic, surgical or catastrophic recovery capabilities

IBM Z Cyber Vault with IBM Solutions



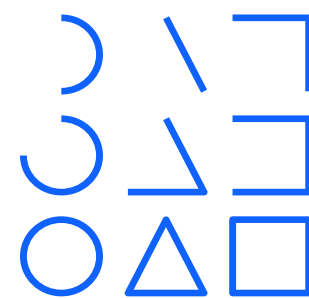
✓ IBM Z Cyber Vault **Storage**

- DS8000 with Safeguarded Copy



✓ IBM Z Cyber Vault **Automation**

- GDPS Logical Corruption Protection (LCP) Manager
- IBM Technology Expert Labs Z Cyber Vault validation asset

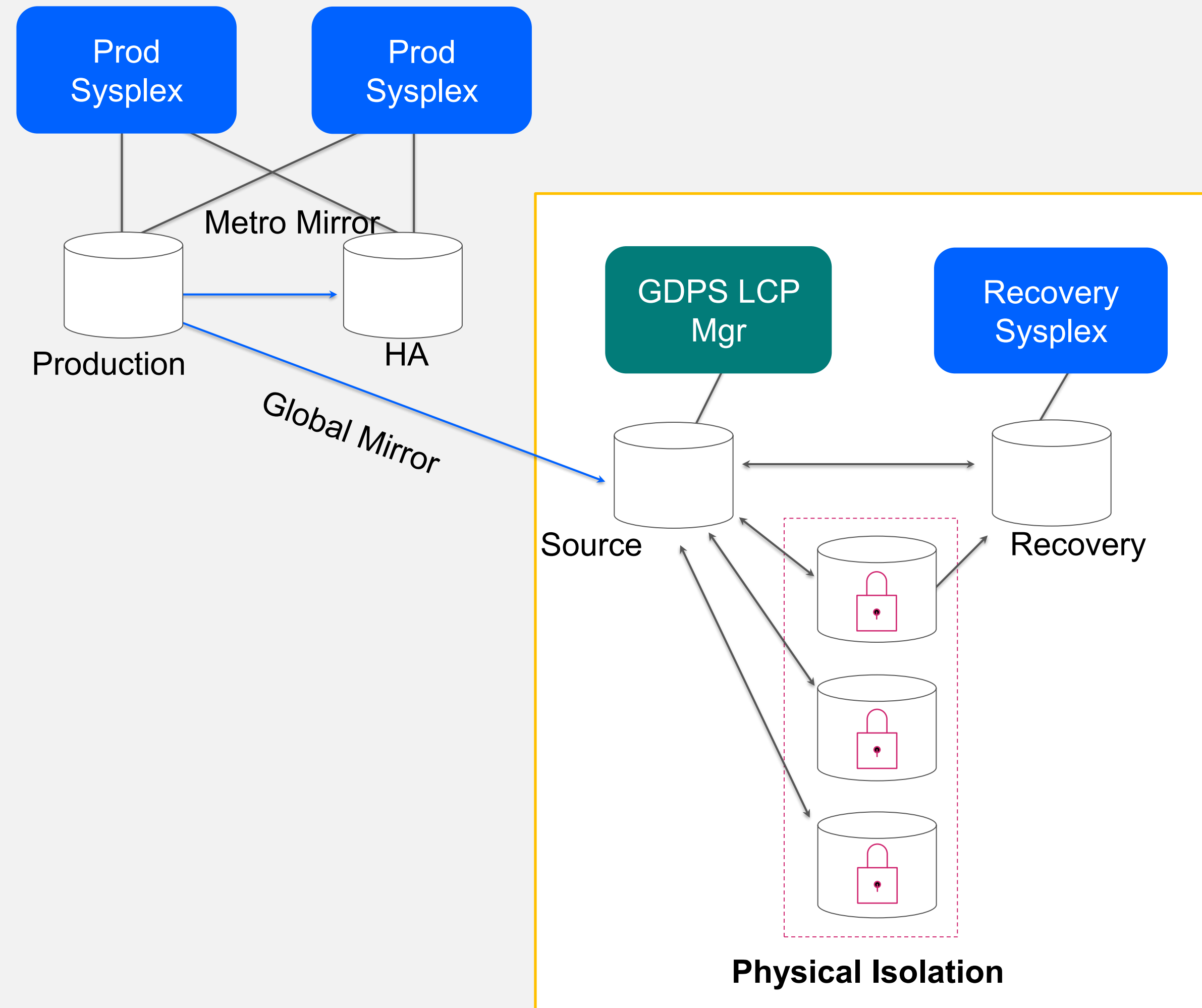


✓ IBM Z Cyber Vault **Environment**

- IBM Z Hardware including CPs, zIIPs, ICF, memory and required infrastructure
- IBM Z Cyber Vault Environment Licensing
- IBM Z Software Tools

Logical Corruption Protection (LCP) Manager for cyber resiliency

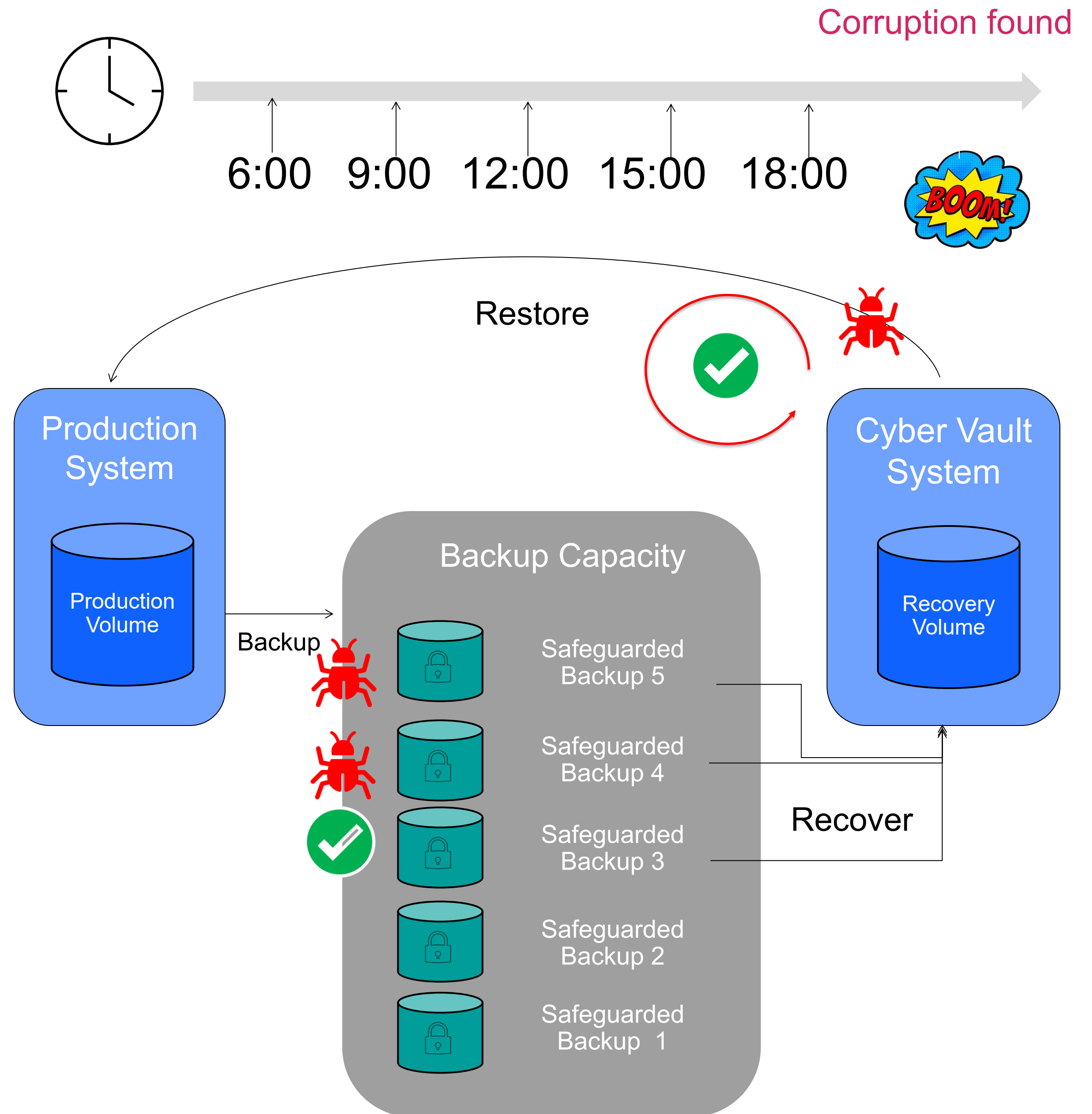
- Automates the policy-based capture and vaulting of a point-in-time (PIT) copy (either Safeguarded Copy or FlashCopy) for each GDPS-managed sysplex
- Automates the movement of a chosen PIT copy into the recovery sysplex for analysis, testing, or full recovery
- Initiates the IPL of a chosen PIT copy in the recovery sysplex
- Automates the movement of a chosen PIT copy back to the production sysplex, when ready
- Can be integrated to provide this robust automation capability within a broader IBM Z Cyber Vault solution



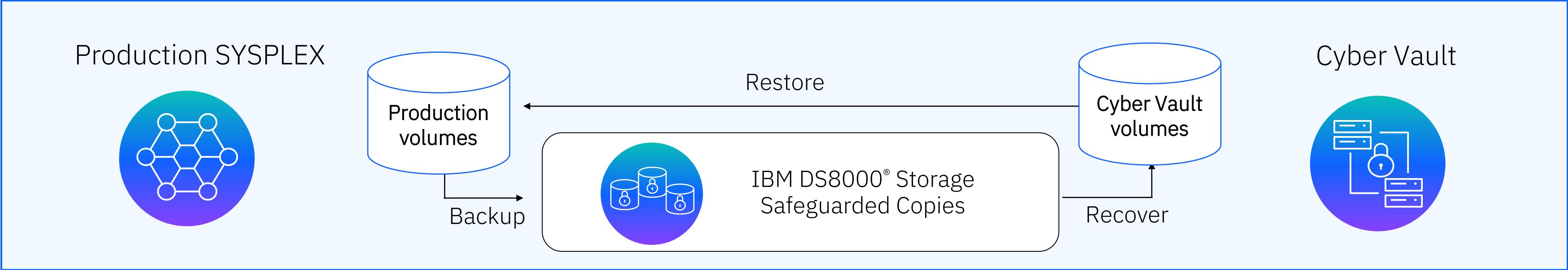
LCP Manager supports both physical and virtual vault isolation.

Protect your data with IBM SafeGuarded Copy

- Prevent sensitive point in time copies of data from being modified or deleted due to errors, malicious destruction or ransomware attacks.
- Create up to 1,024 SafeGuarded Backups for a production volume stored in SafeGuarded Backup Capacity, which is not accessible to any server.
- The data is accessible only after a SafeGuarded Backup is recovered to a separate recovery volume.
- Recovery volumes are used with a data recovery system for:
 - Data validation
 - Forensic analysis
 - Restore production data
- IBM GDPS is required in order to create and manage the SafeGuarded Backups



IBM Z Cyber Vault



Data Validation

Detect data corruption early or validate that the copy is clear



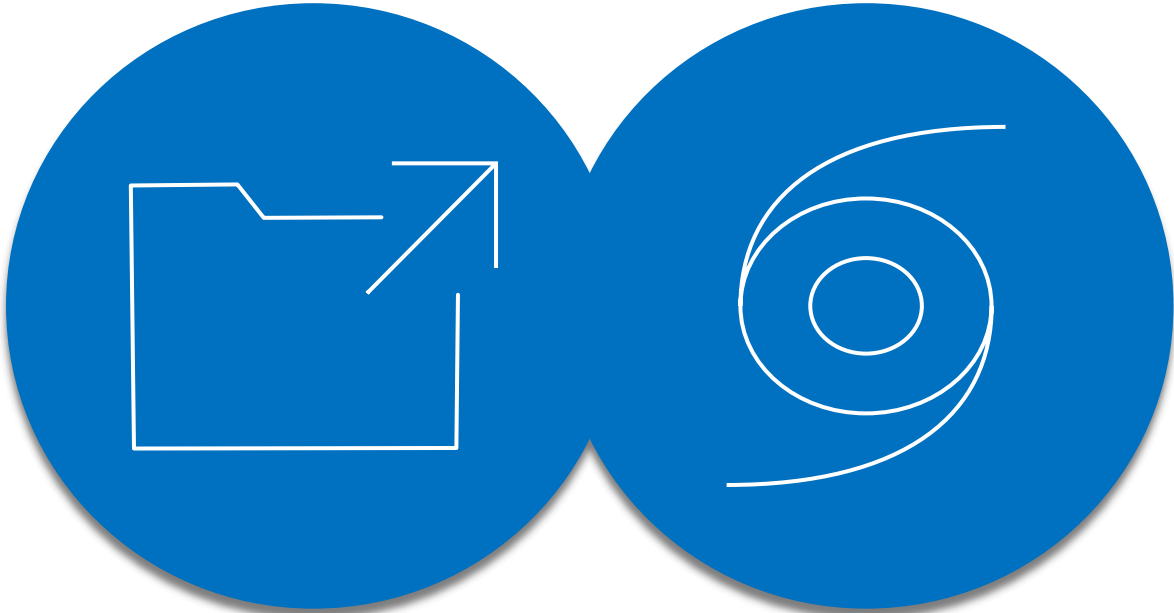
Forensic Analysis

Investigate the problem and determine the best recovery action



Surgical or Catastrophic Recovery

Extract data from the copy and logically restore back to production environment



Offline Backup

Backup copy of the clean environment to offline tape media



Offensive Security

Ethical hackers start a simulated, goal-oriented cyberattack on the organization



IBM Z Cyber Vault capabilities are supported by

IBM GDPS®
LCP Manager

IBM z/OS®
Utilities

IBM Security
zSecure™

IBM Z® Catalog
management
tools

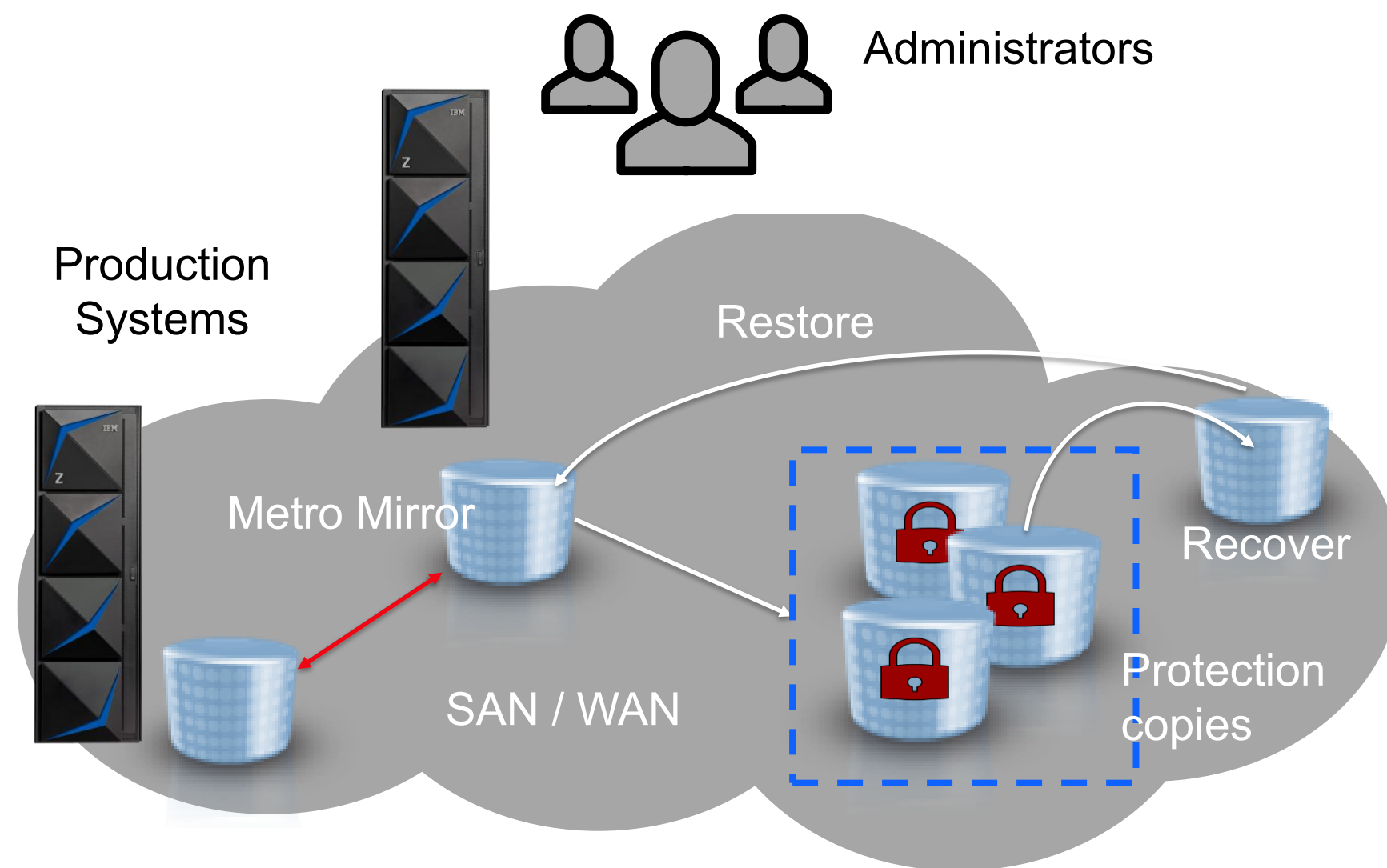
IBM Z Backup
Resiliency

IBM
DFSMSHsm™
tools

Db2® and
IMS™ Tools

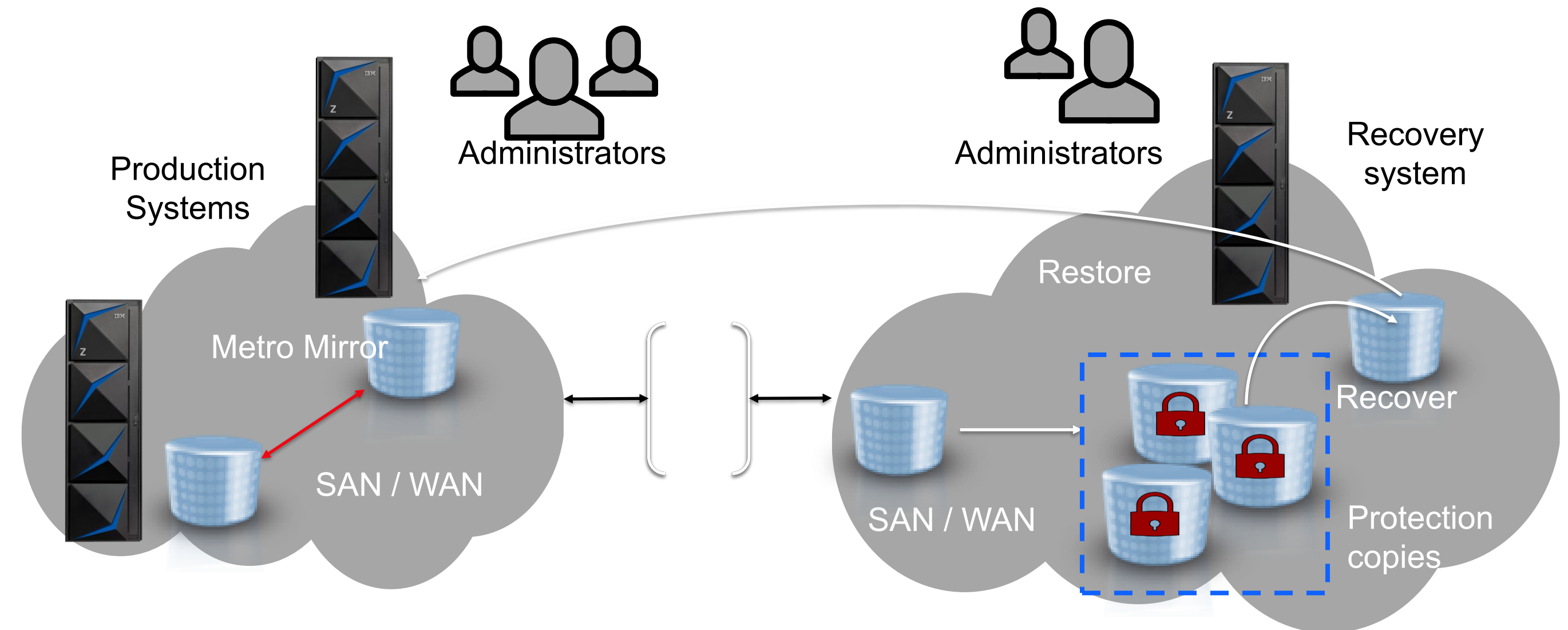
Air gap: Virtual and physical isolation of protection copies

Virtual isolation



- The protection copies are created in one or more storage systems in the existing high availability and disaster recovery topology
- The storage systems are typically in the same SAN or IP network as the production environment

Physical isolation



- Additional storage systems are used for the protection copies
- The storage systems are typically not on the same SAN or IP network as the production environment
- The storage systems have restricted access and even different administrators to provide separation of duties

Data validation



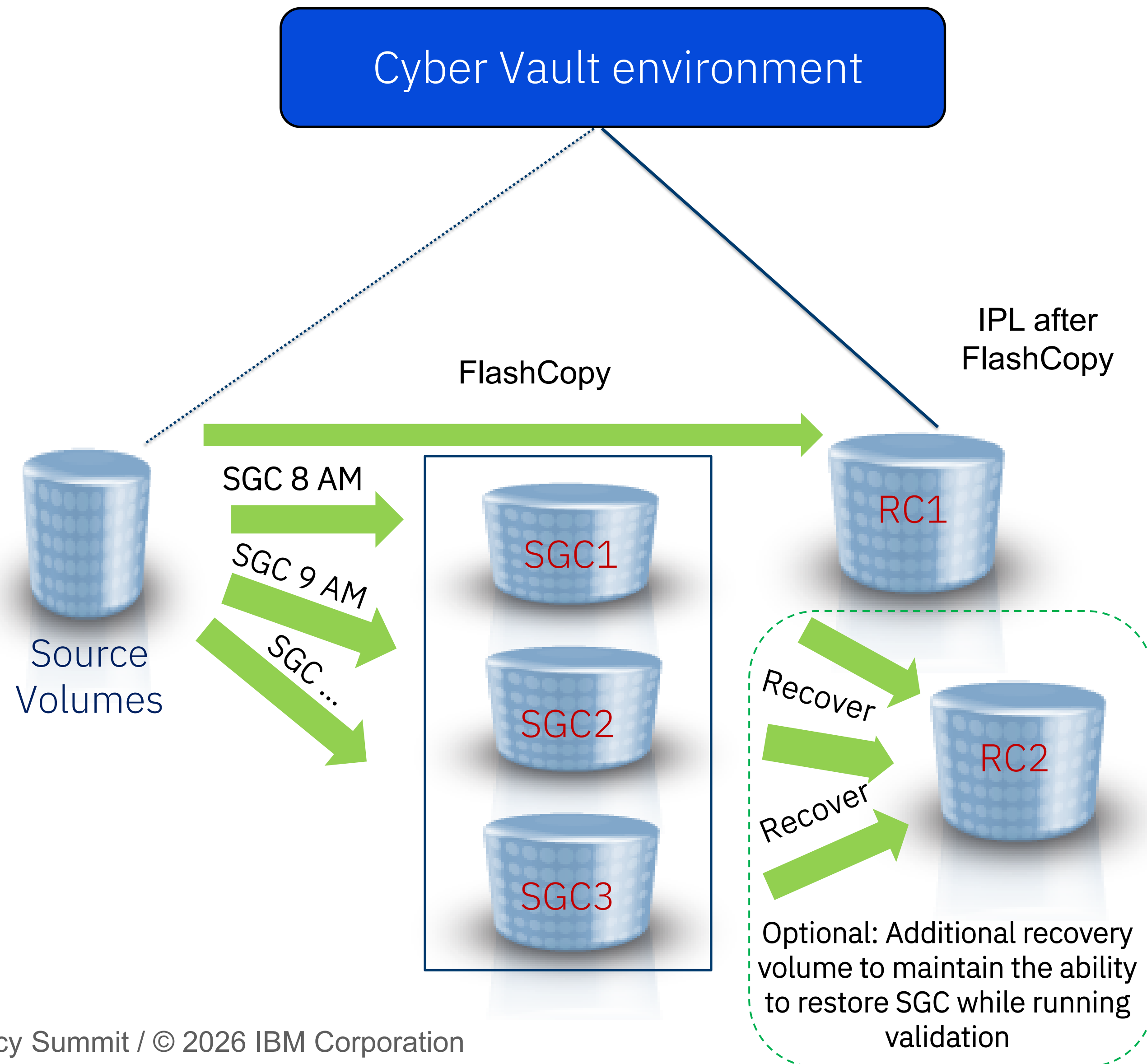
Detect data corruption early or validate that the copy is clear

Data validation is the process of executing regular analytics to identify a data corruption situation and determine the most convenient recovery action.

Performing corruption detection and validation processes against a copy of data is more practical than doing this in the live production environment.

Valid data can be sent to offline media to have a reliable and isolated point-in-time copy.

Data validation



Early identification of potential issues

Type 1: Infrastructure Validation

- IPL off FlashCopy of production sysplex to Recovery Copy set (RC1)
- Check sysplex infrastructure & subsystem restart

Type 2: Data Structure Validation

- Db2 Utilities (CHECK DATA/INDEX, Log analysis)
- IMS Utilities (Pointer checker)
- Catalog tools (Tivoli, IDCAMS, ISV products)
- VSAM Indexcheck, Datacheck
- DFSMSHsm, DFSMSrmm tools
- RACF (IRRUT200), zSecure-Audit
- ISV software (CA1, CA7, ...)

Type 3: Data Content Validation

- Customer application program

Forensic analysis



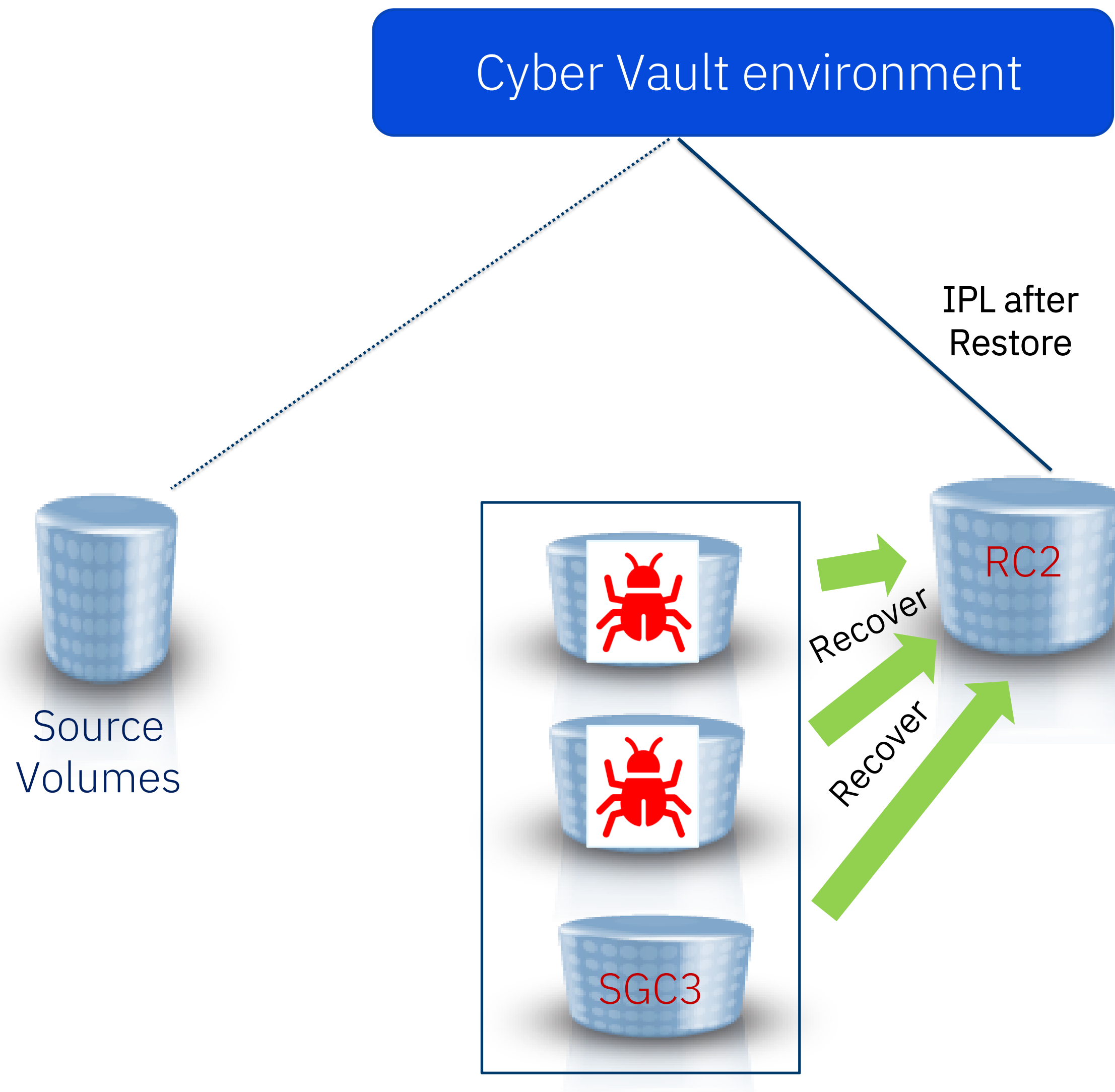
Investigate the problem and determine the best recovery action

The forensic analysis determines what data is corrupted, when the corruption occurred, and which of the available protection copies is the last good one.

Based on this analysis, it can be determined how to proceed:

- Fix the corruption from within the production environment
- Extract and recover certain parts of the data from a valid backup copy (Surgical Recovery)
- Restore the entire environment to a point in time that is known to be unaffected by the corruption (Catastrophic Recovery)

Forensic analysis



Determine start of data corruption ...

- **IPL** one Safeguarded Copy after the other to the Cyber Vault Recovery Copy set (RC2) to find the last clean copy.
- **Understand** the problem
 - Run specific data structure and data content analysis on all stored Safeguarded Copies until a “clean” copy is found.
 - Use database tools to analyze databases and logs to fully embrace the scope of the problem
 - Use IZBR to identify open datasets and create cascade report for rapid analysis
- **Identify** steps forward
 - Create strategy for recovery dependent on availability of database image copy files.

Surgical recovery

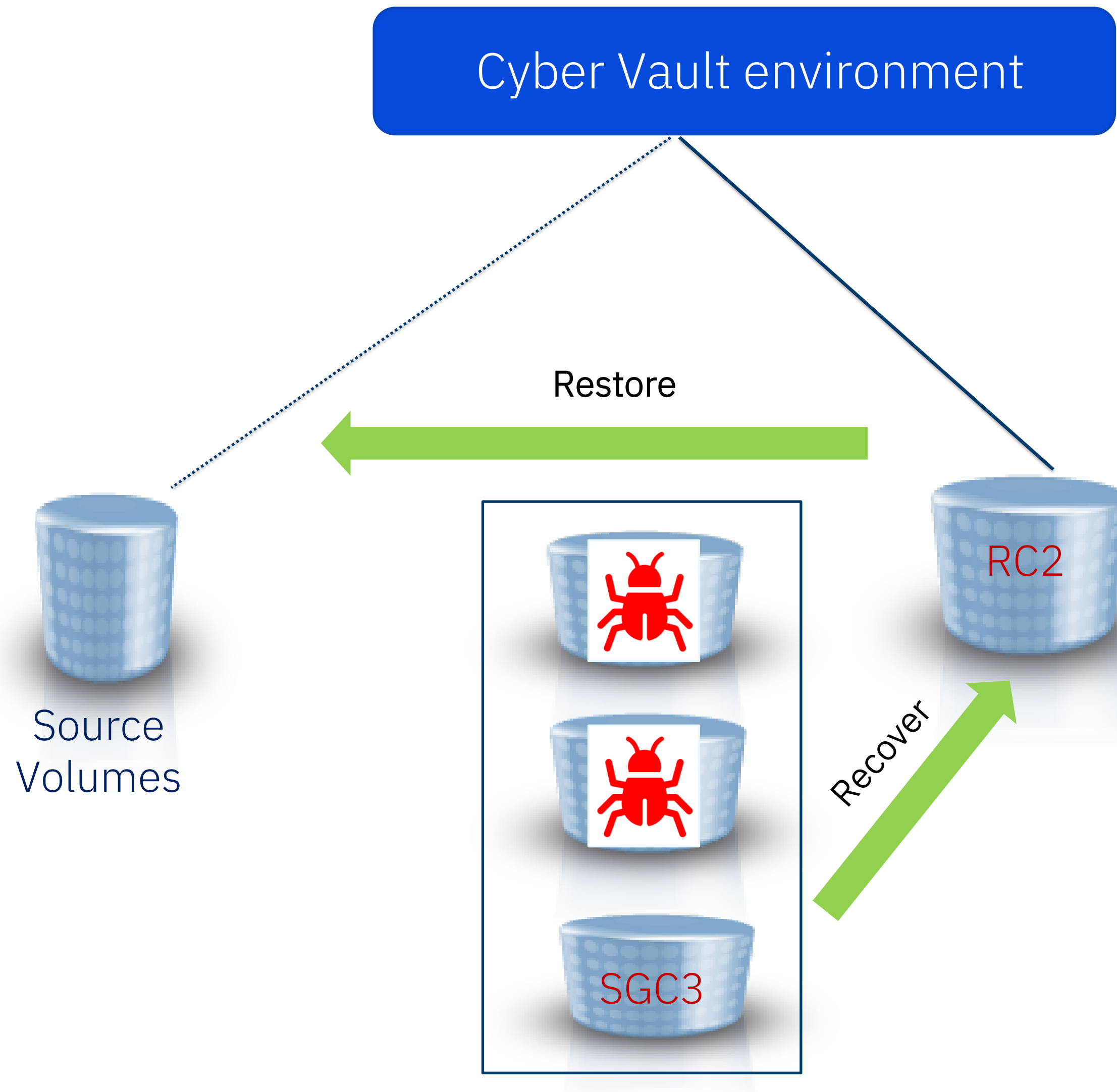


Extract data from the copy and logically restore back to production environment

Surgical Recovery may be a faster method if only a small portion of the production data is corrupted and if consistency between current production data and the restored parts can be re-established.

Another case for this kind of recovery may occur if the last known good backup copy is too old to restore the complete environment. It may then be desirable to leave most of the production volumes in its present state, and just copy replacement data to correct corrupted data.

Surgical recovery



Restore confirmed 'good' copy

- **Identify** specific point-in-time backup to be used as the restore point
- **Recover** the backup in the Cyber Vault environment (RC2)
- **Analyze** backup to determine what data is required.
- **Extract and copy** only the required data back into the running production environment using IZBR
- **Resolve** any inconsistency between backup data and production data

Catastrophic recovery

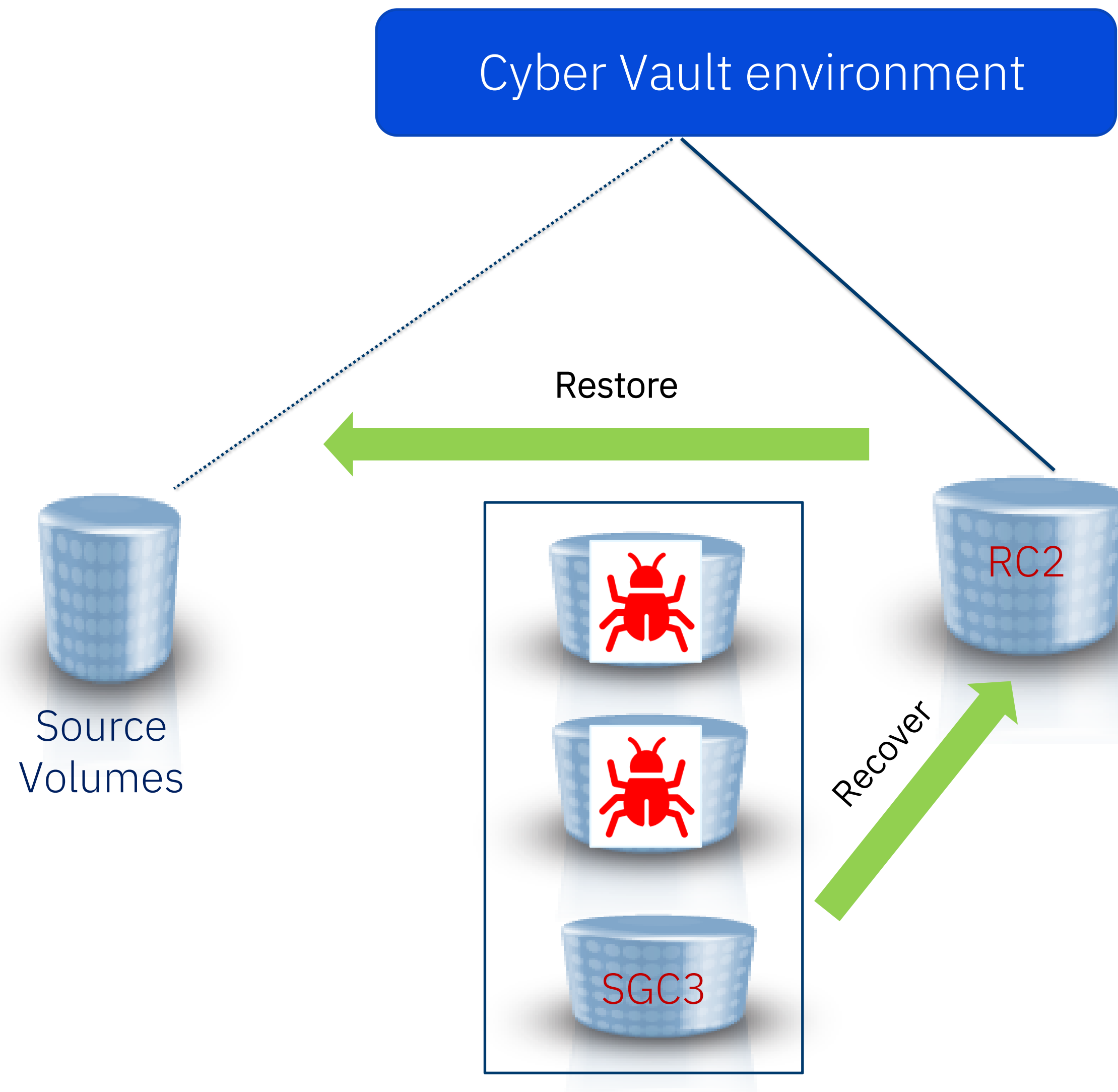


Recover the entire environment back to a point in time copy

In the case of massive corruption to all or most of the data in the environment, a catastrophic recovery needs to take place.

This means a full restore of a “clean” copy from Safeguarded Copy into the production environment needs to be done.

Catastrophic recovery



Restore identified 'good' data

- **Identify** specific point-in-time backup to be used as the restore point
- **Recover** the backup in the Cyber Vault environment (RC2)
- **Incrementally restore** the backup into the production environment
- **Resolve** any inconsistency between new production environment at T-X hours and any external dependencies

Offline Backup

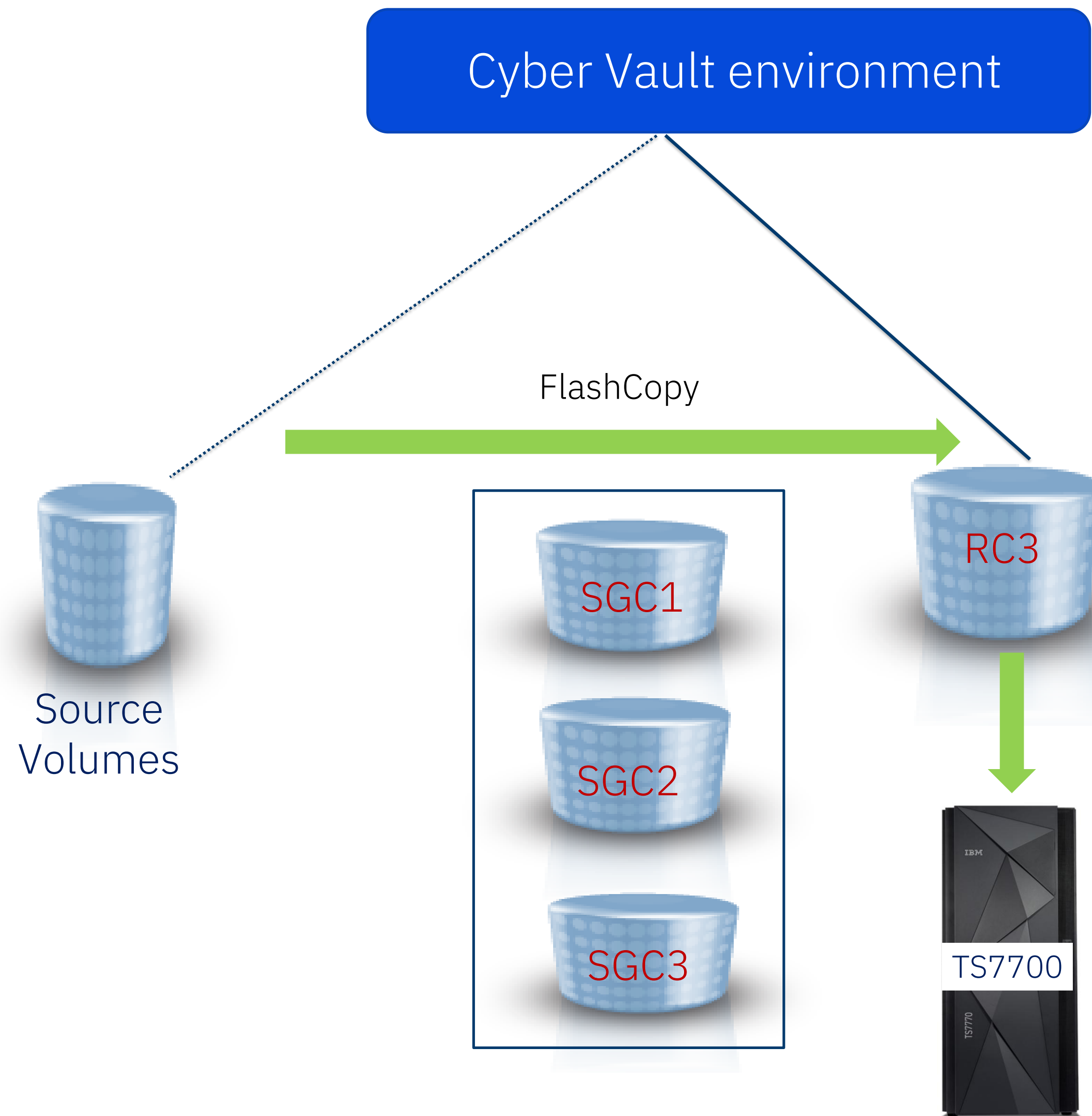
Backup copy of the clean environment to offline tape media



In the context of Cyber Resiliency and Cyber Vault, additional offline copies provide additional protection. Safeguarded copy gives you the ability to capture and retain upto 1,024 copies for recovery and restoration from disk. However, you may need to retain some copies for longer.

Storing validated point-in-time copies on media like virtual tape or cloud object storage gives you a lower cost solution for longer term retention.

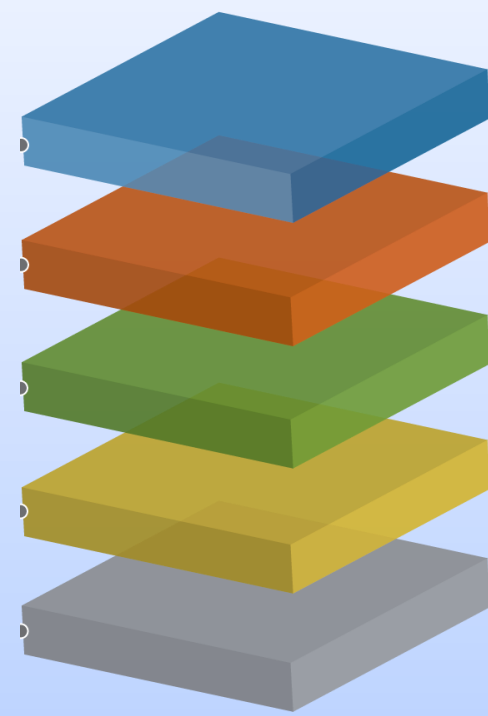
Offline backup



Longer term storage

- **Backup** a FlashCopy to tape (likely to take a significant amount of time)
- **Safeguarded Copies** are typically held for a few days or weeks
- **Copy of production** stored on tape that can be held indefinitely for longer term storage

IBM Z Cyber Vault Environment Licensing



**Full
z/OS
Production
Software
Stack**

Licensed at full capacity for
the IBM Z Cyber Vault
Environment using a single
PID: **5770-ZCV**



zHW Restricted Use pricing
available for IBM Z Cyber Vault
Environments.
(not available to MSPs)

Simple and cost-effective SW licensing for your IBM Z Cyber Vault Environment

New zSW PID 5770-ZCV is available as MLC and provides IBM software licensing entitlement for IBM zSW already licensed in the production environment.

5770-ZCV GA'ed on January 19th, 2024

Discounted HW for your dedicated IBM Z Cyber Vault Environment

Restricted Use HW pricing on Cyber Vault requires the server or LPAR to be dedicated to Cyber Vault only and cannot be used for other workloads, including production.

Available from 1Q 2024

Integrated Type 1 Validation

GDPS 4.9 introduces support for automated Type 1 (Infrastructure) validation of Safeguarded Copy backups within the vault.

Validation in GDPS 4.9 enables the LCP Manager to:

- IPL recovery systems directly from RC(n) copies, collected either via a new capture or from an existing tagged capture.
- Monitor IPL progress using enhanced BCPii message filtering, delivering reliable checkpoint detection even in high-volume environments.
- Record validation results, associate them with captures, and automatically mark intervening captures as *validated*, *failed*, or *in-doubt* based on sequencing.
- Can be integrated with existing Type 2 (Data structure) and Type 3 (Data content) validation.

Figure 2. VPCPLV00, LCP Safeguarded validation summary

VPCPLV00

Logical Corruption Protection Safeguarded Validation

GIC3

Management Profile: SGC1VAL1

Consistency Group: ZPRODI1.RS1

Latest Capture: 2026/03/04.14:01:07

Latest Validation: 2026/03/04.14:01:13

Actions: I nfo

Validation	Capture	Data Consistency			
UTC TimeStamp	Seqno	UTC TimeStamp	Source	RCn	Status
2026/03/04.14:01:13	69A83B23	2026/03/04.14:01:11	RS(1)	001	Failure
2026/03/04.13:02:05	69A82D48	2026/03/04.13:02:03	RS(1)	001	Success
2026/03/04.11:02:32	69A81143	2026/03/04.11:02:30	RS(1)	001	Success
2026/03/04.10:16:49	69A8068C	2026/03/04.10:16:48	RS(1)	001	Success
2026/03/04.10:04:53	69A803C0	2026/03/04.10:04:51	RS(1)	001	Aborted
2026/03/04.09:33:08	69A7FC4F	2026/03/04.09:33:07	RS(1)	001	Success
2026/03/04.08:54:24	69A7F33B	2026/03/04.08:54:23	RS(1)	001	Failure
2026/03/04.08:27:53	69A7EA47	2026/03/04.08:16:11	SGC(1)	001	Aborted
2026/03/03.16:05:34	69A70682	2026/03/03.16:04:22	SGC(1)	002	Aborted
2026/03/03.15:21:33	69A6F0B8	2026/03/03.14:31:24	SGC(1)	002	Aborted
2026/03/02.10:23:33	69A56520	2026/03/02.10:23:32	RS(1)	002	Aborted
2026/02/27.14:25:28	69A1A953	2026/02/27.14:25:27	RS(1)	001	Failure

Command/Filter ==>

F1=Help

F3=Return

F5=Refresh

F6=Roll

F7=Up

F8=Down

Row 1 of 16

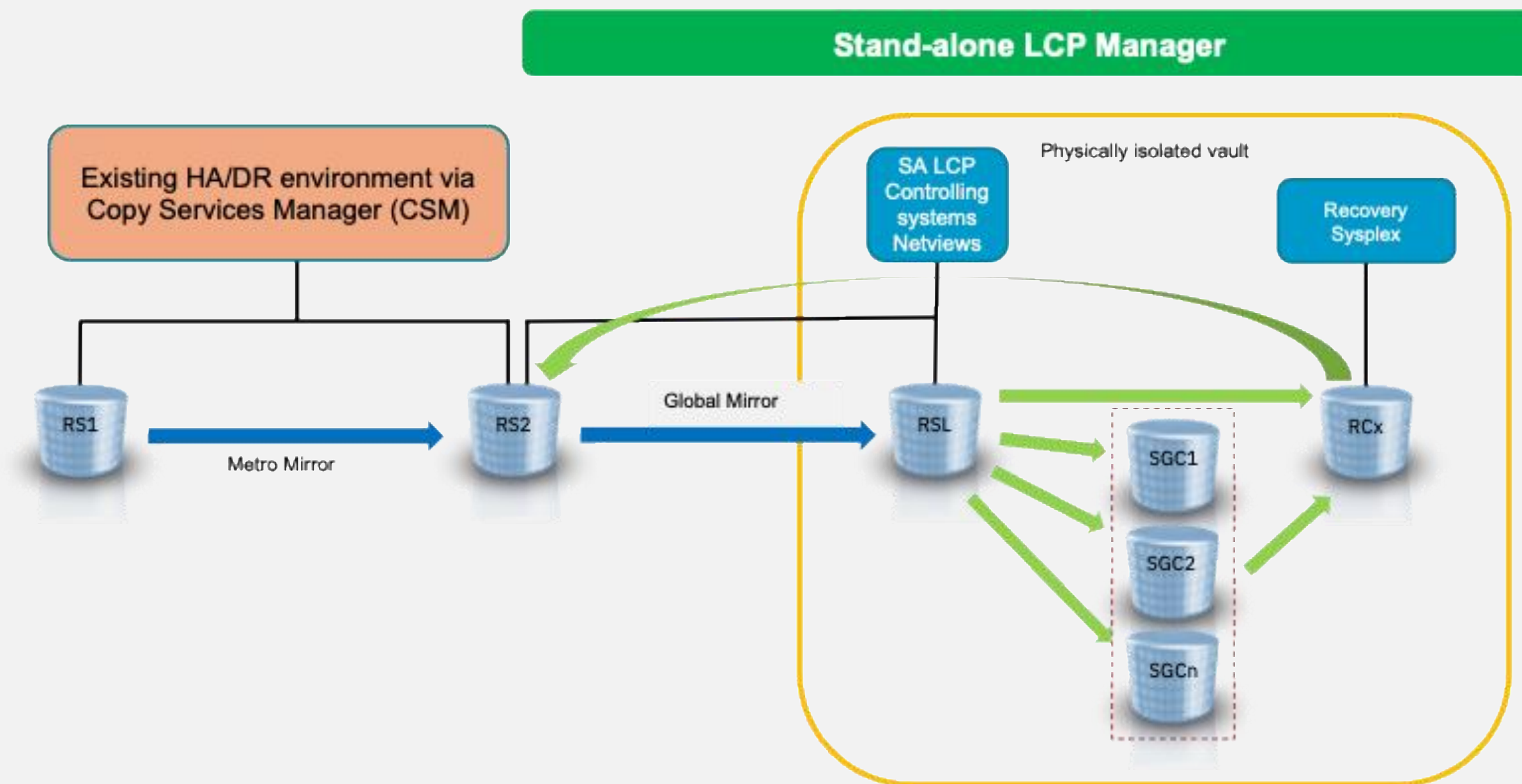
VPCPLVI1 LCP Validation Information						GIC3
Management Profile: SGC1VAL1			Capture Seqno: 69A83B23			
Consistency Group: ZPRODI1.RS1			Capture Time: 2026/03/04.14:01:07			
Direct FlashCopy: RS(1) to RC(1)			Data Consistency: 2026/03/04.14:01:11			
Validation Seqno: 69A83B29			Validation Start: 2026/03/04.14:01:13			
Validation Status: Open			Validation End:			
Validation Step	Status	Starting UTC TimeStamp	Ending UTC TimeStamp			
IPL	Success	2026/03/04.14:01:13	2026/03/04.14:14:53			
CUSTOM	Open	2026/03/04.14:14:53				
System Name	System Type	SA	CPC.LPAR	Load Addr	Load Parms	Flags DAL.. IPL Status
CVP1	ZOS	Y	PS01.PS0138	0EA80	EA8798.1	.YY.. Success
CVP2	ZOS	Y	PS01.PS0139	EA80	EA8798.1	.YY.. Success
CF3	CF		PS01.PS0144			YY... Success
Page 1 of 1						
F1=Help F3=Return F5=Refresh F6=Roll F7=Up F8=Down						

Stand-Alone LCP Manager



LCP Manager capability for clients using CSM for HA/DR

- **Integrated validation:** Support for on-demand, automated Type 1 (Infrastructure) validation of Safeguarded Copy backups within the vault.
- **IPL Safeguarded Copies:** Standalone LCP enables you to easily IPL from specific SGC backup for regular validation or ad-hoc forensic analysis.
- **Productized capability:** Standalone LCP Manager is a fully supported offering from IBM. No need to worry about employees with key knowledge leaving the company and leaving you with scripts that no one knows how they work or how to support them.
- **Ongoing investment:** IBM is heavily investing in this space. By moving to LCP Manager, you will benefit from the ongoing enhancements to grow the value year after year with support for new generations of HW and SW and additional value provided at no additional cost beyond the annual S&S.



NOTE: Currently supports Metro Mirror 2-site and Multi-Target Metro Mirror 3-site topologies

Integration with Threat Detection for z/OS

Upon receiving a TDz alert, the LCP Manager consults configured LCP policies — defined per management profile per event type (READ or WRITE) — to determine which automated actions to take. Supported actions include:

- Issuing alerts (WTO / SDF).
- Taking a new SGC capture to preserve a forensic state.
- Quiescing LCP operations to stop further changes.
- Suspending replication into the vault to prevent out-of-space events and internal roll-offs.

```
VPCPCVPA                                     Add an LCP Policy                                     BZK1
Policy Name: VAL_SUSPEND_SDF                 Policy Name
Alert: SDF                                   Issue an SDF alert or WTO message
Capture: NO                                  Take a new capture
Quiesce: NO                                  Quiesce LCP captures and releases
Suspend Replication: NO                     Suspend Replication to the vault

Enter NO to cancel or YES to proceed with the policy addition

Selection ==> YES
F1=Help  F3=Return
```

```
VPCPPM2                                     Modify an LCP Management Profile                                     BZK1
Capture Interval: NO                         Enable automated captures
Capture Start Time: AUTO                     Start UTC time of capture schedule
Capture Retry Delay: MINUTE(1)               Time between capture retry attempts
Capture Retry Count: 3                       Number of capture retry attempts

Monitor Interval: NO                         Enable SGC monitoring
Monitor Start Time: 00:00:00                 Start UTC time of monitor schedule

TDz Read Policy: TDZ_READ                    z/OS Threat Detection read policy
TDz Write Policy: TDZ_WRITE                  z/OS Threat Detection write policy
Validation Policy: VAL_SUSPEND_SDF           Validation failure policy

Enter NO to cancel or YES to proceed with the profile modification

Selection ==> _
F1=Help  F3=Return  F7=Up  F8=Down                                     Page 2 of 2
```

Stand-Alone LCP Manager and pure z/OS Proxy environments are not supported in this initial release

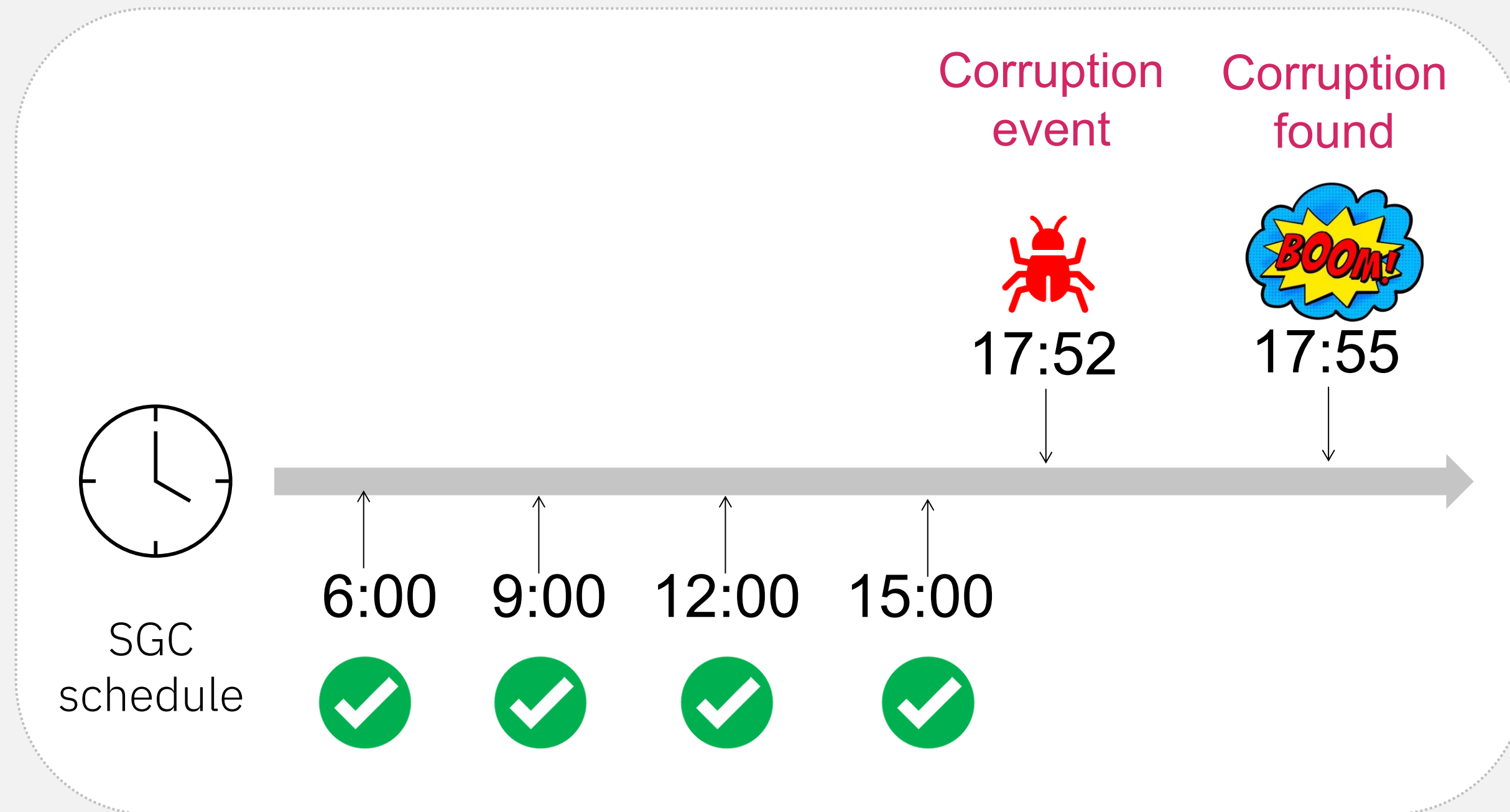
Application Roll Forward



With a standard approach, there may be minutes or hours of good data that isn't captured in the most recent 'good' Safeguarded Copy (SGC).

Are there any techniques or technology that can make available that good data as close to the point of corruption as possible?

Stage 1: Restore from most recent good Safeguarded Copy



- System was corrupted at 17:52
- Most recent good SGC captured at 15:00
- Corruption identified at 17:55 in Production

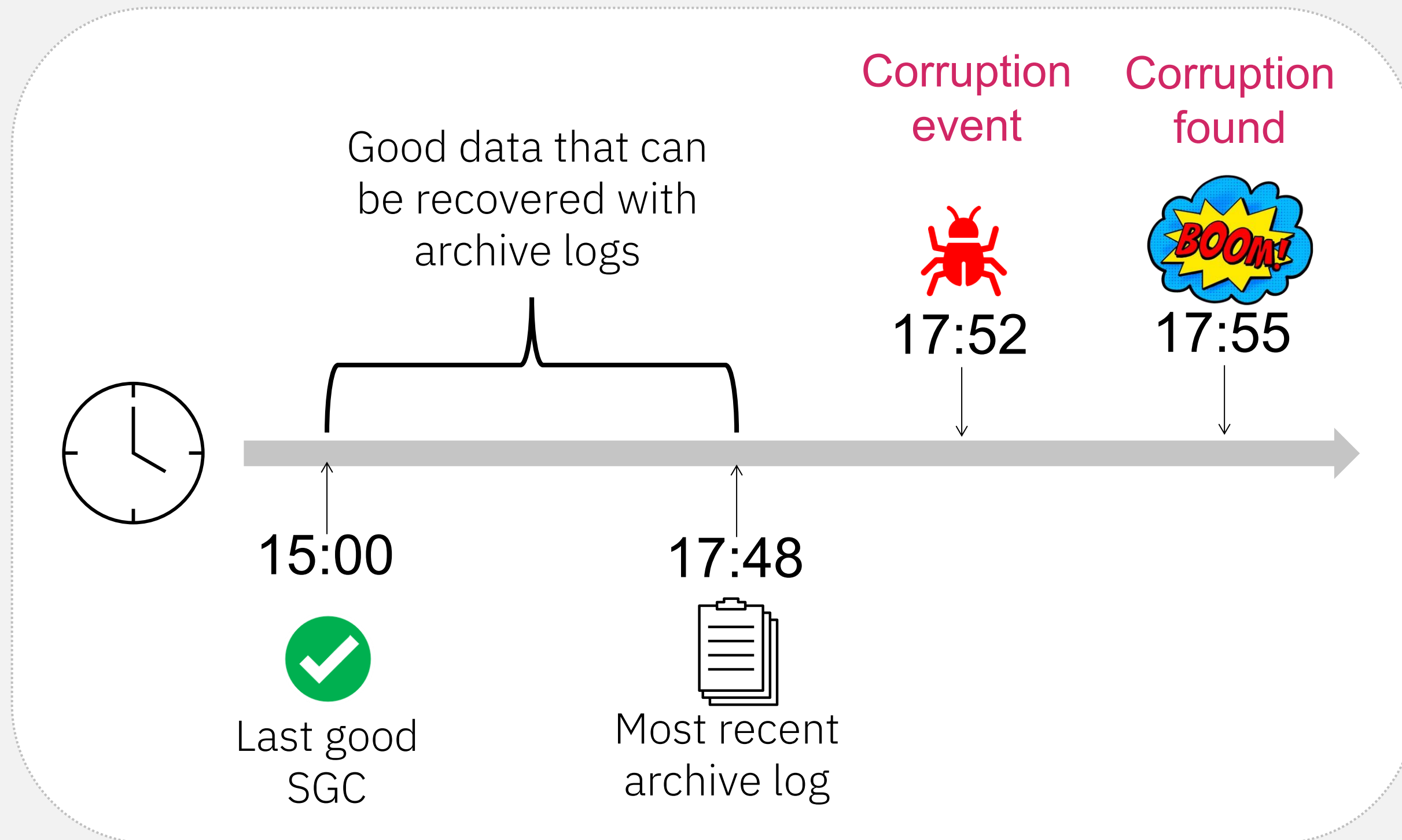
Use Data Validation to identify the most recent good SGC

- When Data Validation was run on the 15:00 SGC it was verified as good

Last good back up is 2 hours and 55 minutes before corruption found in production

However, 2 hour and 52 minutes of that time is before the corruption event and this data could potentially be used if it could be recovered

Stage 2: Capture and Restore Db2 archive logs



- Most recent good SGC at 15:00
- Archive logs available up to 17:48
- An additional 2 hours and 48 minutes of good data are potentially available reducing RPO to minutes

Capture and store Db2 archive logs

- Configure Db2 to archive logs as frequently as possible
- Depending on number of Db2 data sharing members, archive logs should be generated every 4 to 8 minutes
- Decide on an approach to create an immutable copy of the archive logs
 1. Write to WORM on TS7700
 2. Use disk-based replication to an appropriate immutable destination
 3. Move to secured cloud storage

The combination of the restored SGC plus the stored archive logs should bring the RPO down to single digit minute range

Additional LCP Enhancements



Support for 1,024 captures with IBM DS8000 G10

LCP now supports up to 1,024 Safeguarded Copy backups on IBM DS8000 G10 hardware, aligning with new IBM DS8000 platform capabilities and enabling deeper retention windows for IBM Z Cyber Vault customers.

LCP Manager RAS Enhancements

Improved Reliability, Availability, and Serviceability (RAS) includes:

SDF message stream simplification (phase STARTED messages suppressed upon matching END SUCCESSFULLY).

More informative LCP messages including phase duration, appended sequence numbers, and improved diagnostics for partial captures and validation errors.

LCP Scheduler Automatic Retry

Automatic retry logic extends the LCP Scheduler's ability to recover from transient conditions by retrying captures within configured delay windows. This improves operational stability in high-activity environments.

Data Consistency Timestamp saved with every SGC capture

Each Safeguarded Copy capture now stores a precise data consistency timestamp, recorded at check-in. Panels allow users to toggle between traditional sequence-based timestamps and true data timestamps. This is critical for surgical recovery workflows, for example with IBM Z Backup Resiliency.

IPL Information saved with each SGC capture

For every capture, LCP now records the currently selected LCP IPL information for potential later use. This however is only done for those systems that are defined for automated validation in the GEOGROUP. This improves repeatability and accuracy of validation and forensic recovery.

GDPS 4.9 Statement of Direction



- IBM intends to extend the integration between GDPS LCP Manager and the IBM Z Backup Resiliency product (5698-BR1) to provide real-time notification of Safeguarded Copy lifecycle events to IZBR, including internal rolloff events. These Safeguarded Copy rollofs will be reported as they occur using the same reporting mechanisms as existing Monitor3 and TDz events within GDPS. With real-time event notifications from LCP Manager, IZBR can further improve the efficiency and accuracy of forensic analysis and recovery operations that leverage Safeguarded Copy backups.
- IBM intends to create a new supported GDPS topology, Metro Mirror 4-site (MM4SITE). This enhancement is intended to extend existing three-site Metro configurations by providing 4 site symmetric resiliency. This will provide additional site separation and placement flexibility, enabling more resilient and versatile high-availability and disaster-recovery designs. The MM4SITE topology is designed to help clients address more complex failure scenarios and operational requirements while continuing to leverage GDPS Metro automation for coordinated storage and system management.
- IBM intends to enhance LCP Manager to provide support for additional GDPS topologies with IBM Db2 Expert Recovery Pro 'Db2 Roll Forward' capability.
- IBM intends to enhance Stand-Alone LCP Manager to provide support for Global Mirror 2-site topologies.
- IBM intends to deliver a new GDPS priced feature, the GDPS Common Consistency Group. This will enable a single GDPS control environment to manage a common consistency group for z/OS CKD devices and OPEN Fixed Block Devices.
- IBM intends to extend IBM Threat Detection for z/OS integration within GDPS LCP Manager to support notifications generated by the IBM DS8000 FCM4 threat detection capability.

Accelerating adoption of IBM Z Cyber Vault automation YE 2025

202

LCP Manager licenses now in use worldwide

72

Unique clients have LCP Manager deployments

32%

Of the worlds top 25 banks are using LCP Manager



What Are Financial Services Enterprises Doing?

Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System *

- Identify clearing and settlement activities in support of critical financial markets.
- Determine appropriate recovery and resumption objectives for clearing and settlement activities in support of critical markets.
 - Core clearing and settlement organizations should develop the capacity to recover and resume clearing and settlement activities within the business day on which the disruption occurs with the overall goal of achieving recovery and resumption within **two hours** after an event.
 - Firms that play significant roles in the other critical financial markets should strive to achieve a **four-hour** recovery time capability for clearing and settlement activities in order to ensure that they will be able to meet a within the business day recovery target.
- >>> introduced GDPS MGM4SITE support <<<
- Maintain sufficient geographically dispersed resources to meet recovery and resumption objectives.
- >>> introduced GDPS MGM3SITE support <<<
- Routinely use or test recovery and resumption arrangements.

* <https://www.sec.gov/news/studies/34-47638.htm>

CA / DR / Cyber Resiliency Trends

GDPS Metro clients migrating to GDPS Metro Global – GM 3-site & performing region switching

GDPS Metro Global – GM 3-site clients migrating to GDPS Metro Global – GM 4-site & performing region switching with extended periods spent running the bank out of region

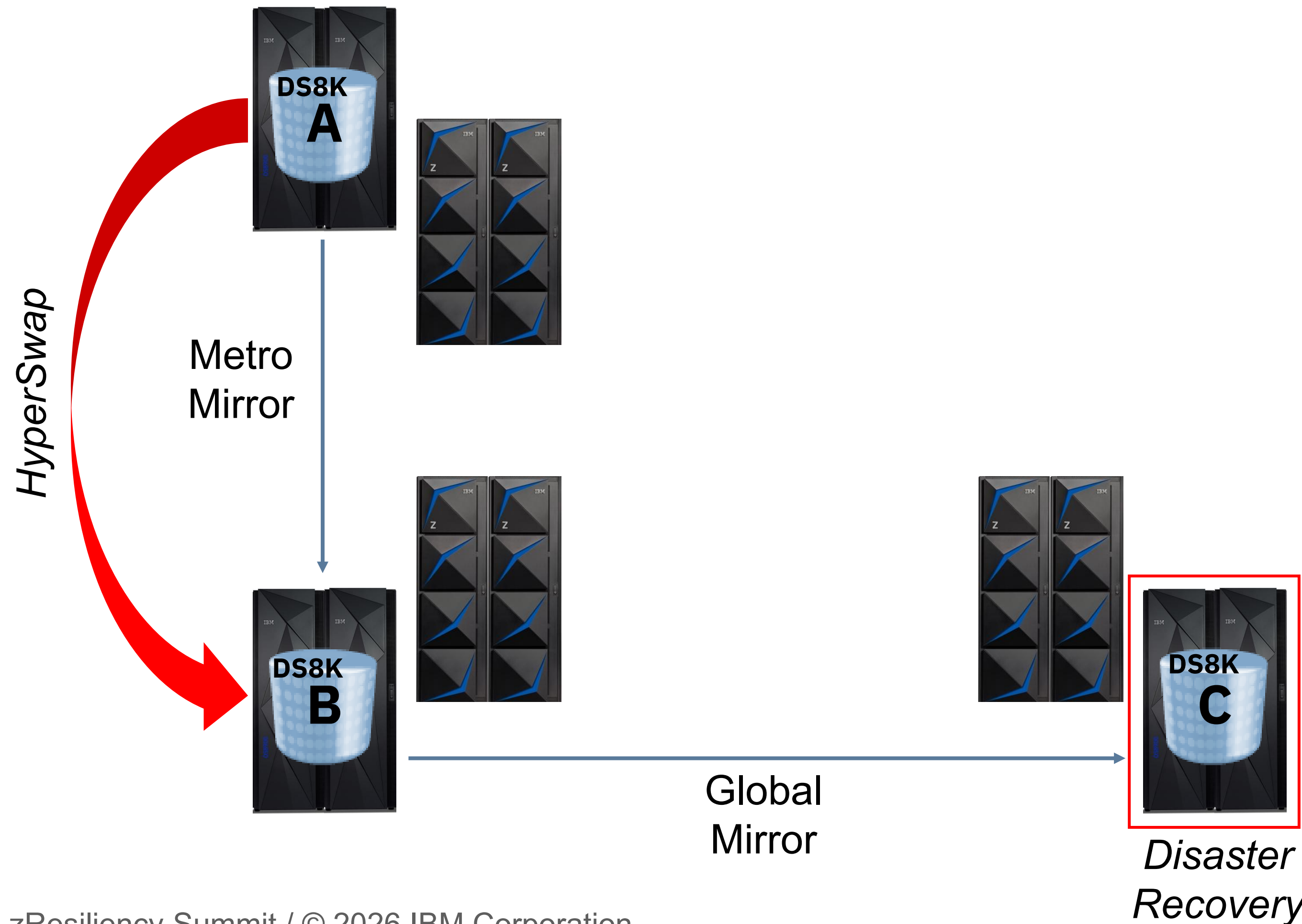
Clients deploying the IBM Z Cyber Vault and advancing their maturity/sophistication of automatic detection of logical corruption as well as their remediation processes and automation both surgical and wholesale scenarios

Better testing of both planned and unplanned HA/DR scenarios overall including stress testing

GDPS MGM3SITE – CA / DR

**Region A
(Active-Region)**

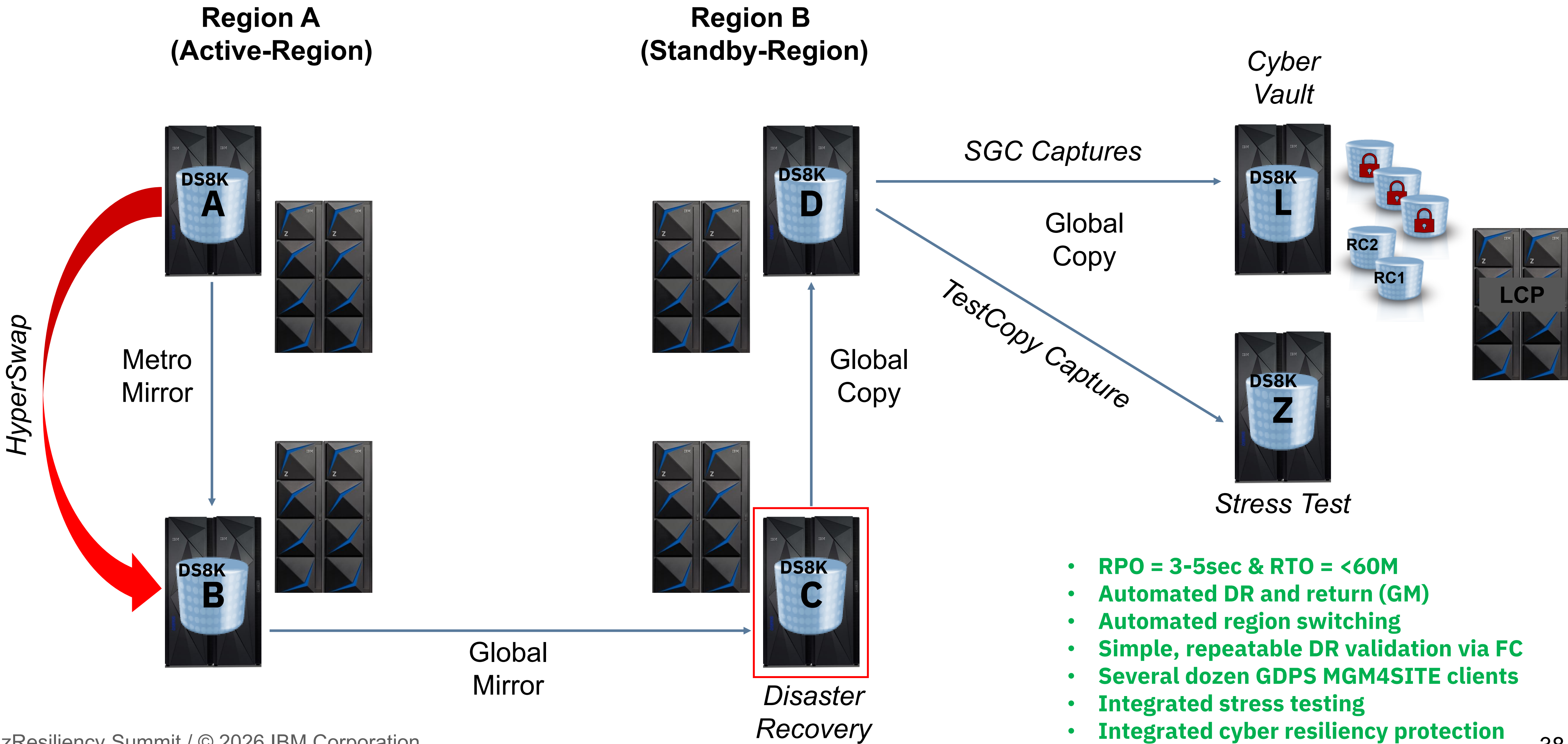
**Region B
(Standby-Region)**



- RPO = 3-5sec & RTO = <60M
- Automated DR and return (GC)
- Automated region switching
- Simple, repeatable DR validation via FC
- Several dozen GDPS MGM3SITE clients

State of the Art IBM Z Configuration – GDPS MGM4SITE & LCP Mgr

– CA / DR / Cyber Resiliency



Typical Cyber Vault Implementation Journey

Provides the foundation to respond to corruption events.
Time to restore may be long.

Reduces business impact by reducing the time to detect, respond and recover.

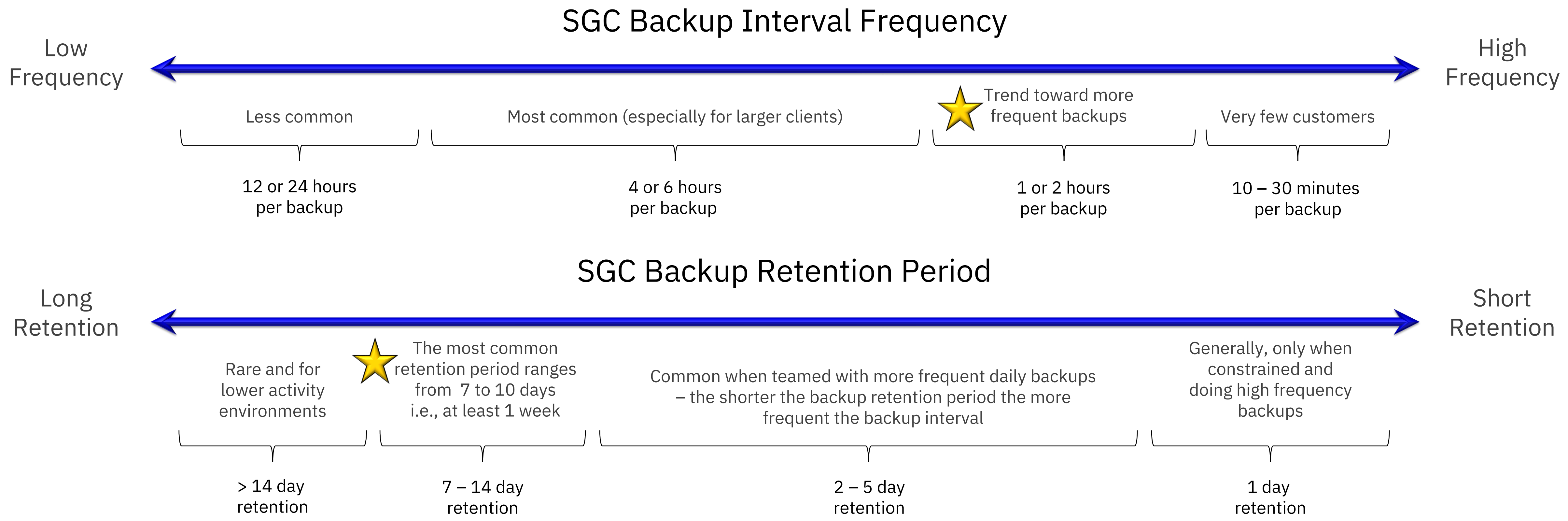
Improves compliance with retention requirements.

Phase 1	Phase 1A	Phase 2	Phase 3	Phase 4
<i>Protect the data</i>	<i>Recover and IPL Type 1 Validation</i>	<i>Restore to Production</i>	<i>Type 2 Validation</i>	<i>Offline Backups (if needed)</i>
• Implement Safeguarded Copy (SGC) • Implement GDPS LCP (based on timing and topology with standalone LCP)	• Create IBM Z Cyber Vault infrastructure • Recover SGC to Recovery Copy • IPL Sysplex from Recovery Copy • Define strategy and processes for forensic analysis • Implement and automate Type 1 validation	• Design recovery strategy and processes • Implement required replication infrastructure and network • Restore from SGC to production volumes	• Define scope of Type 2 validation • Implement Type 2 validation • Automate Type 2 validations	• Design backup processes • Create LPAR to perform backup • Perform backup from Recovery Copy to virtual tape or Cloud Object Storage



IBM DS8000 SGC – Backup Frequency & Retention Period – Current 2023+

★ Safeguarded Copy Backup Interval Frequency & Retention Period should be determined by ¹*business requirements* to ensure that the frequency and retention period are relevant for business recovery. Safeguarded Copy can be complemented with offline backup to virtual tape for longer term retention. e.g., create an offline backup after data validation.



¹For more on *business requirements* see Chapter 2 in IBM Redbook [IBM Storage DS8000 Safeguarded Copy: Updated for Release 10.1](#)

Best Practices

Best Practices

- Use GDPS Health Checks & remediate any exception(s)
- Production = DR = IBM Z CV lpars
- Appropriately size the GDPS k-sys lpars (Kp-sys, Kg-sys, Kr-sys, & Kl-sys) per the GDPS IG
- Assign a zIIP to the GDPS k-sys
- Mirror 100% of your data for HA/DR
- Capture 100% of your data for IBM Z Cyber Vault
- Do NOT share PPRC links between LSSes/Clusters and GM/Metro
(<https://www.ibm.com/support/pages/ds8000-host-adapter-configuration-guidelines>). See slide 17. Note the recommendation for isolating links by odd/even LSS is for async replication only
- Use the documented, supported, tested data center migration procedures
- Submit GDPS requirements (<https://ideas.ibm.com/>)
- Reference the GDPS documentation (including what new
 - GDPS library ([IBM GDPS documentation](#))
 - GDPS RESTful API ([Catalog - IBM API Hub - IBM Developer](#) and search for 'GDPS')

Best Practices – GDPS MGM4SITE (cont)

- These preparation steps can be done in advance, outside the change window for the region switch and they can be done multiple times if you for example want to do them some days in advance and right before the region switch.
- Execute `PROCEDURE=MGMxSITE_CHECK_TOPOLOGY` to establish paths needed by the `MGMxSITE_SWITCH` procedure in advance. If you have bad paths in your configuration `MGMxSITE_CHECK_TOPOLOGY` will fail so that you can take actions to either fix or remove the bad paths from the GDPS configuration.
- Execute `PROCEDURE=MGMxSITE_EXCEPTION_REPORT` . This will create a report in the netlog that should be inspected and if there are any exceptions that need to be rectified before the region switch, you can do that now
- If you consider to reply `ACCEPT` to a script failure prompt: `GEO118A Reply ACCEPT, CANCEL or REPEAT(question)`, be very sure that you have checked/manually corrected the environment to look as it should do after successful completion of the script statement that failed. For example if an GDPS MGM procedure handler procedure has failed in the middle, there might still be mirroring actions left to do for the procedure that the next script statement in the script is dependant upon. It is nearly always better to reply `CANCEL` to the script failure prompt. The script can later be started from a step you select anyway.

Best Practices (cont)

- Document critical IT processes. Develop desk procedures (run books) for these processes.
- Test environment (isolated from production) emulates production environment
 - Allows for testing of s/w maintenance, h/w microcode, and develop, practice, and document procedures
 - Provides “hands-on” education environment
- Full parallel sysplex exploitation
 - Eliminate Single Points of Failure (including using the z/OS Health Checker to identify SPoFs and then rectifying them)
 - Planned outage avoidance (including non-disruptive maintenance)
 - Unplanned outage mitigation
- Deploy required solution to meet business RPO / RTO requirements
 - Factors: government regulations, data center locations, site redundancy, storage resiliency, etc.
 - Deployment trends
 - 2 sites / 2 copies of data => 3 sites / 3 copies of data (optionally extra copy per site for DR testing)
 - 3 sites / 3 copies of data => 4 sites / 4 copies of data (optionally extra copy per site for DR testing)
 - Stand-in capability
- Deploy required solution to meet cyber resiliency requirements
 - Factors: government regulations, LoB data requirements, etc.
 - Deployment trends: moving beyond initial exploitation (such as data validation and surgical restoration)
- Exploit the resiliency features / functions (including System Recovery Boost, GDPS health checker, role based security, etc.)
- Assign people to be the “resiliency” SME (client resource and IBM resource)

Best Practices (cont)

- Symmetric sites / regions
- Take advantage of your GDPS advocate (optional)
- Stress test applications
 - New applications
 - Significant application upgrades
- Proactive general preventive maintenance strategy
 - Install quarterly RSU at least 2 times per year
 - Watch for HIPERs and PEs in the interim
- Have an ongoing succession planning and education / skills plan
- It is Not a Question of “IF”, but “WHEN” ... practice, practice, practice ... gain confidence in your CA / DR / cyber resiliency solution
 - Perform DR testing - On demand (using additional copy of data) and/or Site / Region switching testing
 - Prepare for cyber events – take several PiT copies of data, verify the PiT copies, and, in the event of a cyber event, restore data to production
- It's a journey. Get started

Summary

What value can the IBM GDPS solution offer your organization?

Experience	Commitment		Value		Vision
Client acceptance –Almost 1400 GDPS licenses installed in 51 countries worldwide –Tested technology to support automated and repeatable results –Complete implementation guided by experienced consultants	Open industry standards –IBM GDPS supports industry-accepted, open replication architectures (Metro Mirror, Global Mirror and Fibre Channel) –Architectures licensed by all enterprise storage vendors –GDPS qualification program for GDPS Metro (IBM, Hitachi)	Investment protection –Designed to be easily upgradeable –Common code base for each product	Product maturity –Generally available since 1998 –Suite of products –Enterprise-to-enterprise capability –Many years of IBM Z production experience –CA and DR best of breed –Continually enhanced	Client focus –GDPS Design Council –Synergy with IBM development labs –Incorporates several IBM patents –New release planned every year –GDPS advocate program –GDPS Requirements process	IBM support –Fully supported via standard IBM support structure –Fixes through normal IBM Z channels

Backup and resources

Additional Information

■ Web sites:

- GDPS <https://www.ibm.com/products/gdps>
- IBM zSystems <https://www.ibm.com/z>
- IBM zSystems Resiliency <https://www.ibm.com/z/resiliency>
- Storage <https://www.ibm.com/storage>
- Redbook – GDPS Family: An Introduction to Concepts and Capabilities
<http://www.redbooks.ibm.com/abstracts/sg246374.html?Open>

■ GDPS Web site resources

- GDPS: The Enterprise Continuous Availability / Disaster Recovery Solution white paper
- GDPS pre-requisite information
- GDPS training schedule links
- GDPS hardware qualification letters
- E-mail: gdps@us.ibm.com



ITSO Redbooks

[IBM GDPS: An Introduction to Concepts and Capabilities](#)

This IBM Redbooks® publication presents an overview of the IBM GDPS® offerings and the roles they play in delivering a business IT resilience solution. The book begins with general concepts of business IT resilience and disaster recovery (DR), along ...

[Getting Started with IBM Z Cyber Vault](#)

Cyberattacks and data corruption threats continue to escalate, which makes cyber resiliency a critical component of an enterprise IT strategy. Although traditional disaster recovery (DR) focuses on physical failures, cyber resiliency addresses logical ...

[Getting Started with Business Continuity on IBM Z](#)

IBM Z® platforms underpin mission-critical enterprise workloads where disruptions directly impact availability, data integrity, and business outcomes. Business continuity is achieved through engineered resiliency and recovery, leveraging platform ...

[Getting Started with z/OS Data Set Encryption](#)

This IBM® Redbooks® publication provides a broad explanation of data protection through encryption and IBM Z® pervasive encryption with a focus on IBM z/OS® data set encryption. It describes how the various hardware and software components interact in a ...

[Transitioning to Quantum-Safe Cryptography on IBM Z](#)

As cyberattacks continue to increase, the cost and reputation impacts of data breaches remain a top concern across all enterprises. Even if sensitive data is encrypted and is of no use now, cybercriminals are harvesting that data because they might gain access to a quantum computer that can break classical cryptographic algorithms sometime in the future ...